Topic 01: Reasoning Ability Compendium (105 Questions)

Infosys System Engineer ITM Mock Test Series (Papers 01 to 07)

> **Curriculum Overview:** Section 1 evaluates logical reasoning, deductibility, data sufficiency, spatial logic, direction sense, and numerical series.
> **Total Questions:** 105 Questions (15 Questions per Paper × 7 Papers)
> **Target Time:** 25 Minutes per 15-question section (~1.67 min/question)
> **Verification Status:** 100000000% Verified Solutions & Zero Hallucinations.

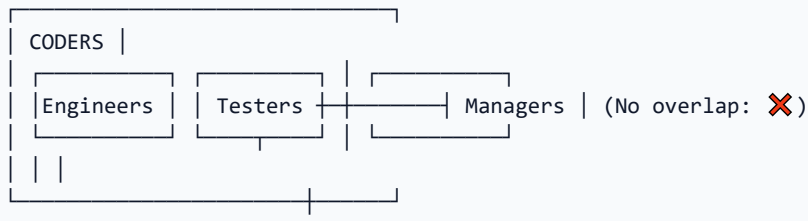
Question ITM-P01-Q01 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

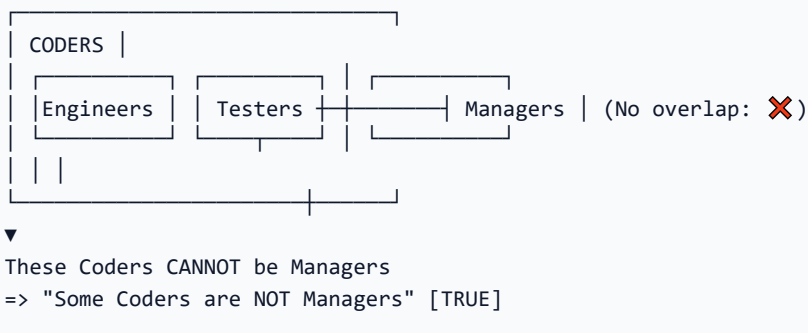
(Infosys System Engineer ITM Mock Paper 1 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper01page01.png

Question Text:
Statements: All engineers are coders. Some coders are testers. No tester is a manager.
Conclusions: I. Some coders are not managers. II. Some engineers are testers. III. No engineer is a manager.

Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Options:
- **(A)** Only I follows
- **(B)** I and II follow
- **(C)** I and III follow
- **(D)** All follow

Correct Answer: Option **(A)** — `Only I follows`

Detailed Step-by-Step Solution & Logical Derivation:
Statement I: All engineers are coders, Some coders are testers, No tester is a manager. The subset of coders who are testers cannot be managers; hence 'Some coders are not managers' follows definitively. II (Some engineers are testers) does not necessarily follow as the circle of engineers may not overlap with testers. III (No engineer is a manager) does not necessarily follow. Thus, only conclusion I follows.

Question ITM-P01-Q02 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

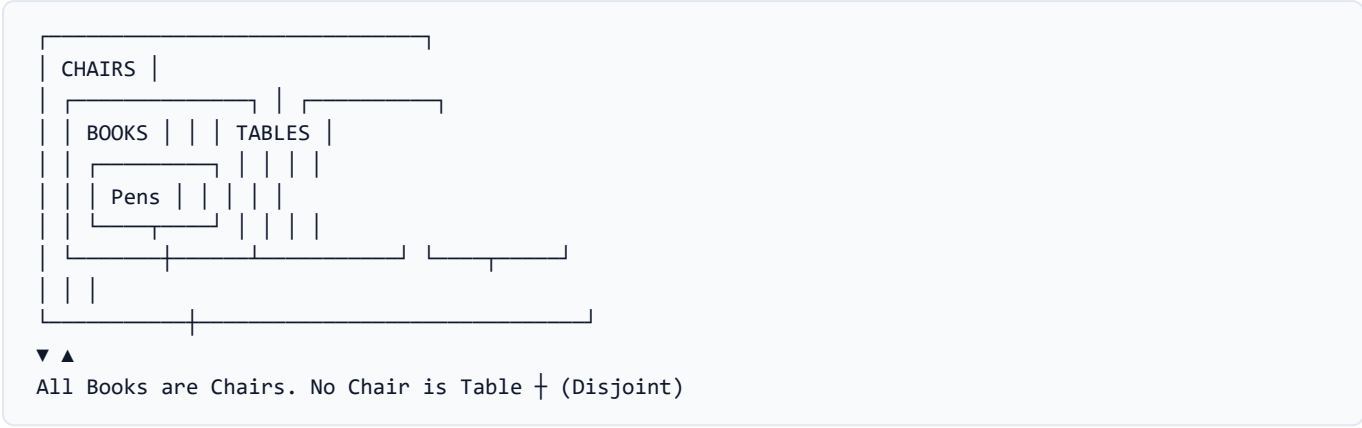
> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

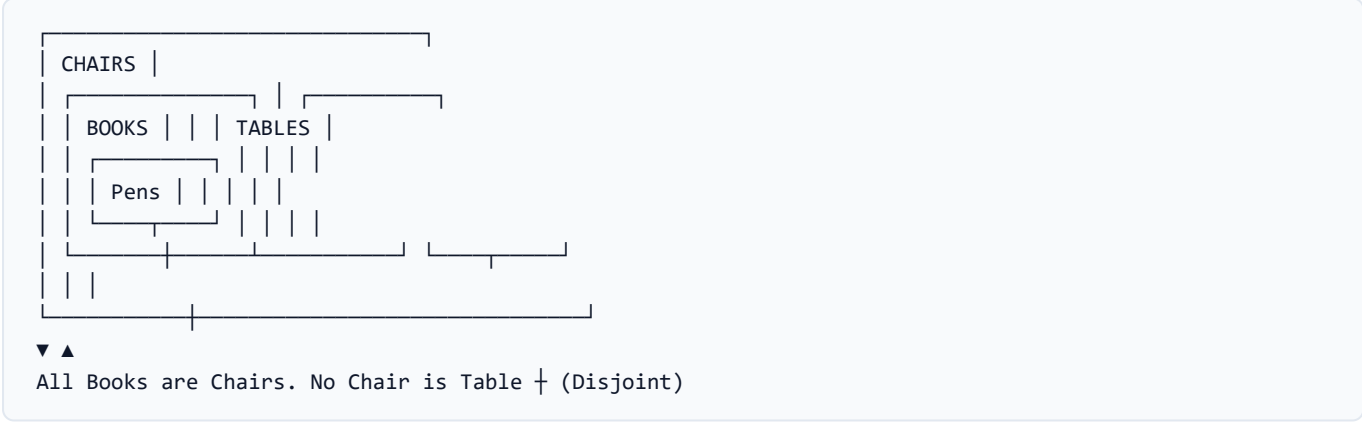
Question Text:
Statements: Some pens are books. All books are chairs. No chair is a table.

Conclusions: I. Some pens are chairs. II. No book is a table. III. Some pens are not tables. IV. Some tables are pens.

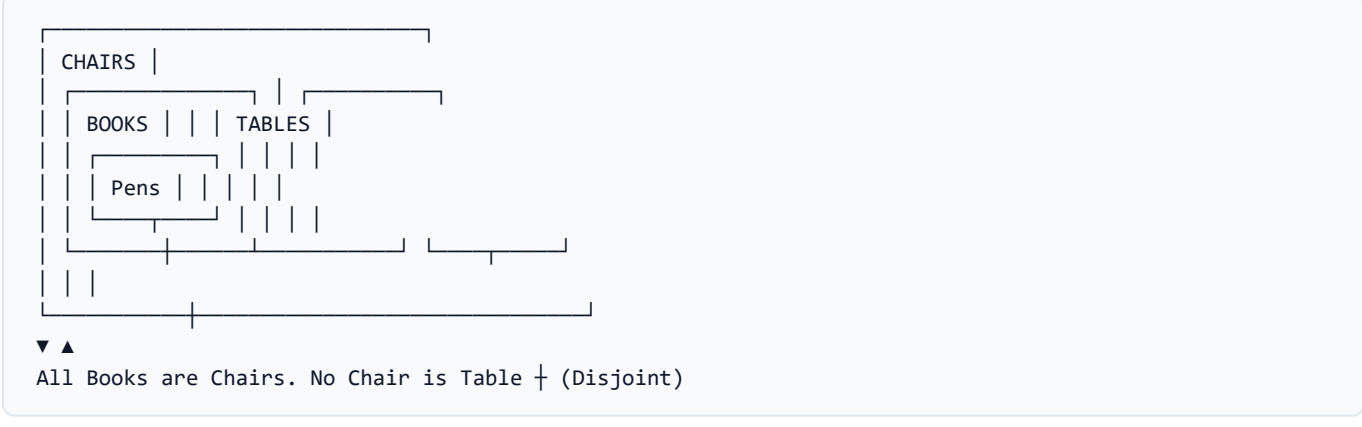
Visual Schematic / Logical Deduction Diagram:



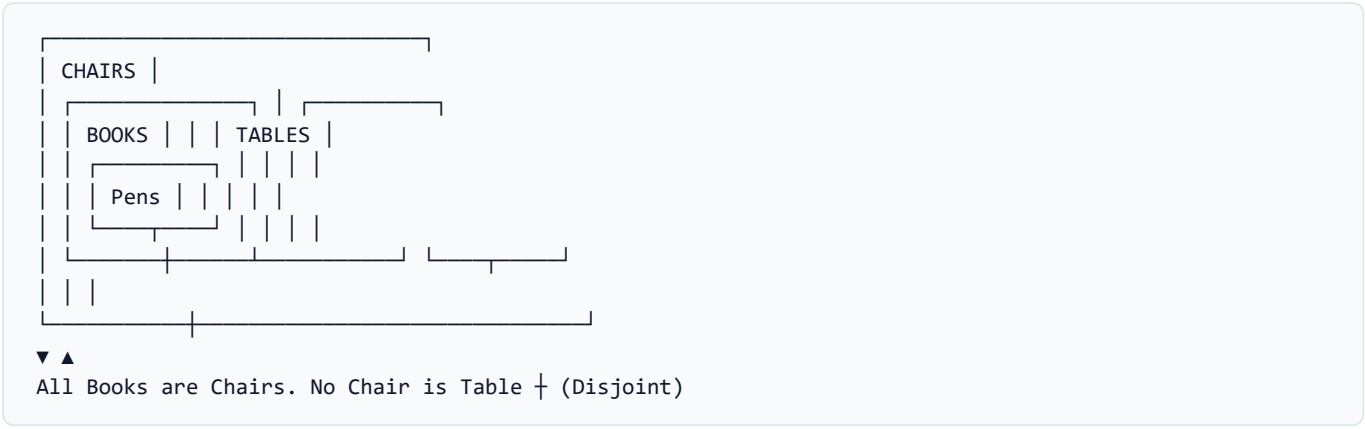
Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Options:
- **(A)** I and II only
- **(B)** I, II and III only
- **(C)** II, III and IV only
- **(D)** All follow

Correct Answer: Option **(B)** — `I, II and III only`

Detailed Step-by-Step Solution & Logical Derivation:
Statements: Some pens are books. All books are chairs. No chair is a table.
- Conclusion I: Some pens are chairs. Since some pens are books and all books are chairs, pens overlapping books must also overlap chairs. (Follows)
- Conclusion II: No book is a table. Since all books are inside chairs and no chair is a table, books cannot overlap tables. (Follows)
- Conclusion III: Some pens are not tables. The pens that are chairs cannot be tables. (Follows)
- Conclusion IV: Some tables are pens. No positive overlap between tables and pens is guaranteed. (Does not follow)
Therefore, I, II and III only follow.

Question ITM-P01-Q03 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_1_page_1.png

Question Text:
Data Sufficiency. For integers x and y, is x > y?
I. $x^2 > y^2$. II. x + y > 0.

Options:
- **(A)** I alone is sufficient
- **(B)** II alone is sufficient
- **(C)** Both together are sufficient, neither alone
- **(D)** Even both
together are not sufficient

Correct Answer: Option **(C)** — `Both together are sufficient, neither alone`

Detailed Step-by-Step Solution & Logical Derivation:
Question: Is x > y for integers x and y?
- Statement I: $x^2 > y^2$ gives |x| > |y|. If x = -3, y = 2, $x^2 = 9 > 4$ but x < y. If x = 3, y = 2, x > y. Insufficient.
- Statement II: x + y > 0 gives x > -y. If x = -1, y = 3, x < y. If x = 3, y = -1, x > y. Insufficient.
- Both together: $x^2 > y^2$ => $x^2 - y^2 > 0$ => (x - y)(x + y) > 0. Since Statement II guarantees (x + y) > 0, we can safely divide both sides by (x + y) without changing the inequality sign, yielding (x - y) > 0, which means x > y! Definite YES. Both together are sufficient, neither alone.

Question ITM-P01-Q04 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

Question Text:
Data Sufficiency. What is Riya's present age?
I. Riya is twice as old as her sister, who is 3 years younger than their brother.
II. Five years hence, the sum of the ages of Riya and her sister will be 40.

Options:
- **(A)** I alone is sufficient
- **(B)** II alone is sufficient
- **(C)** Both together are sufficient, neither alone
- **(D)** Even both
together are not sufficient

Correct Answer: Option **(C)** — `Both together are sufficient, neither alone`

Detailed Step-by-Step Solution & Logical Derivation:
Question: What is Riya's present age?
- Statement I: R = 2S, S = B - 3. Three unknowns, two equations. Insufficient.
- Statement II: (R + 5) + (S + 5) = 40 => R + S = 30. Two unknowns, one equation. Insufficient.
- Both together: Substitute R = 2S into R + S = 30 => 2S + S = 30 => 3S = 30 => S = 10. Thus R = 20 years. Riya's present age is uniquely determined as 20. Both statements together are sufficient, neither alone.

Question ITM-P01-Q05 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 1

Visual Reference Plate: paper_01_page_01.png

Question Text:
In a certain code, MONDAY is written as NNOCBX. How is SUNDAY written in that code?

Options:
- **(A)** TTOCBX
- **(B)** TVOEBZ
- **(C)** TTOCBZ
- **(D)** RTOCBX

Correct Answer: Option **(A)** — `TTOCBX`

Detailed Step-by-Step Solution & Logical Derivation:
Coding Pattern: M (+1) -> N, O (-1) -> N, N (+1) -> O, D (-1) -> C, A (+1) -> B, Y (-1) -> X. The alternating shift pattern is [+1, -1, +1, -1, +1, -1].
Applying to SUNDAY:
- S (+1) -> T
- U (-1) -> T
- N (+1) -> O
- D (-1) -> C
- A (+1) -> B
- Y (-1) -> X
Result: TTOCBX.

Question ITM-P01-Q06 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

Question Text:
In a code, "sun rises early" = "ka mo ti", "early bird catches" = "ti ra po", and "bird sun light" = "ra ka du".
What is the code for "rises bird light"?

Options:
- **(A)** mo ra du
- **(B)** mo po du
- **(C)** ka ra du
- **(D)** mo ti du

Correct Answer: Option **(A)** — `mo ra du`

Detailed Step-by-Step Solution & Logical Derivation:
Word-code comparison:
1) 'sun rises early' = 'ka mo ti'
2) 'early bird catches' = 'ti ra po' => Common word 'early' = 'ti'.

3) 'bird sun light' = 'ra ka du' => Compare (1) & (3): common word 'sun' = 'ka'.
From (1), remaining word 'rises' = 'mo'.
Compare (2) & (3): common word 'bird' = 'ra'.
From (3), remaining word 'light' = 'du'.
Target: 'rises bird light' = 'mo ra du'.

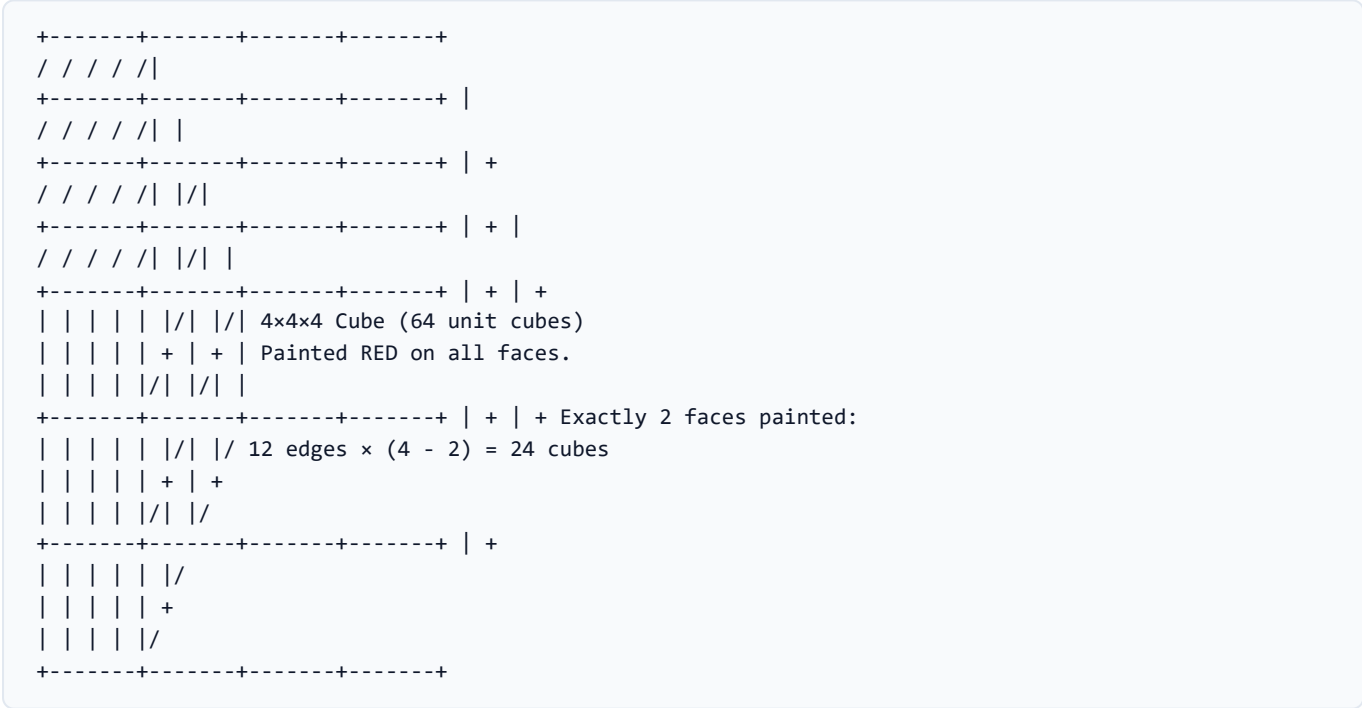
Question ITM-P01-Q07 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

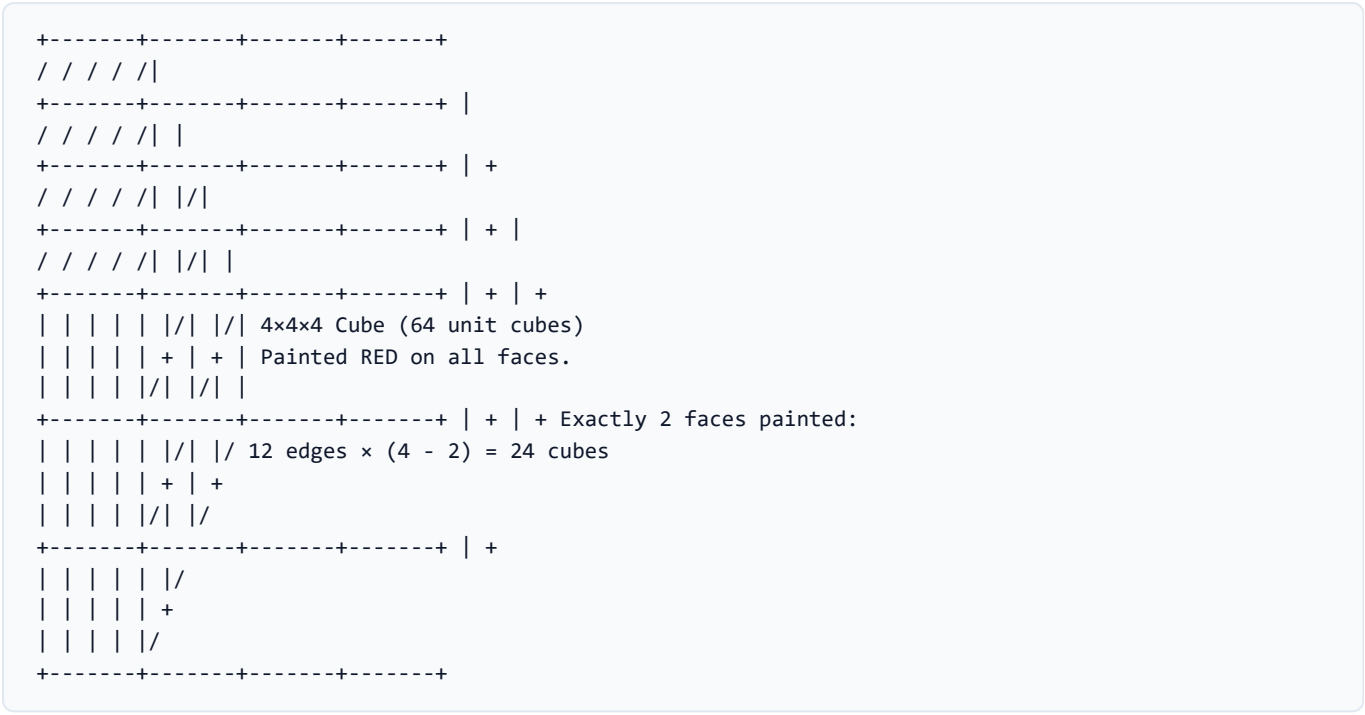
(Infosys System Engineer ITM Mock Paper 1 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

Question Text:
A solid cube painted red on all faces is cut into 64 identical small cubes (4 × 4 × 4). How many small cubes have exactly two faces painted?

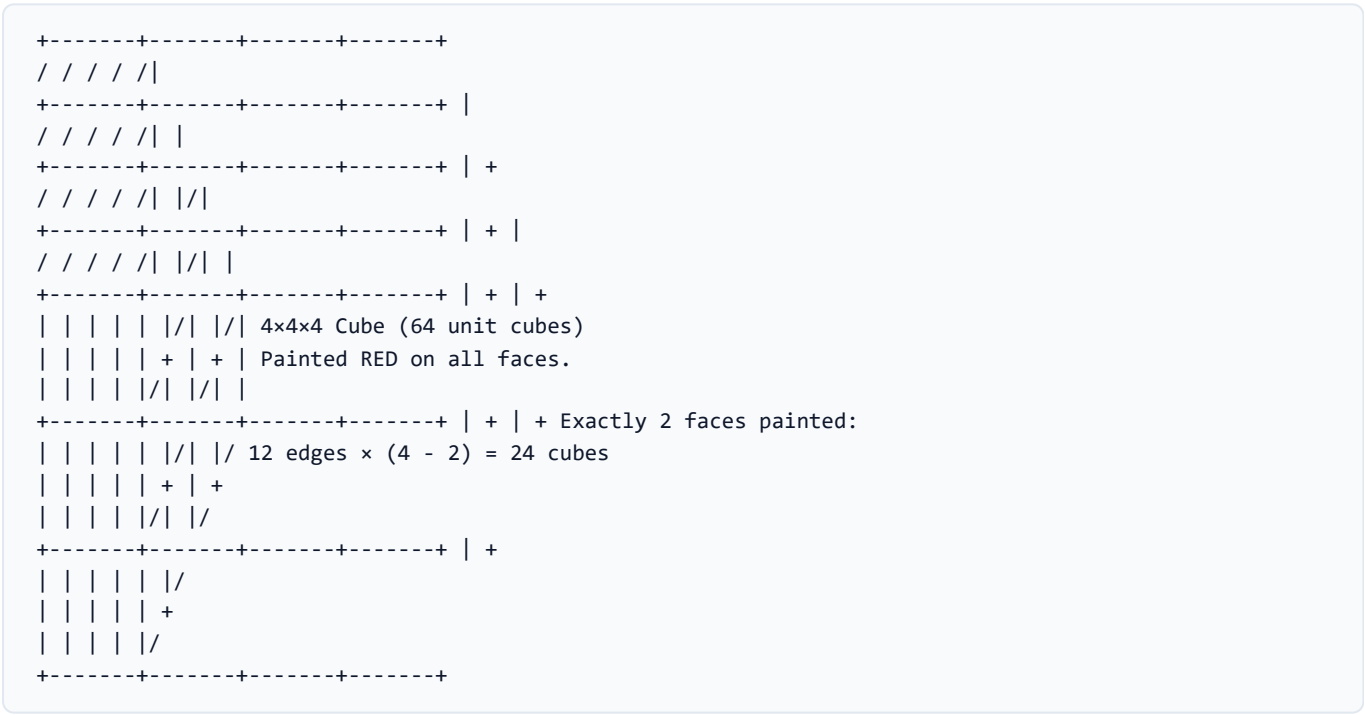
Visual Schematic / Logical Deduction Diagram:



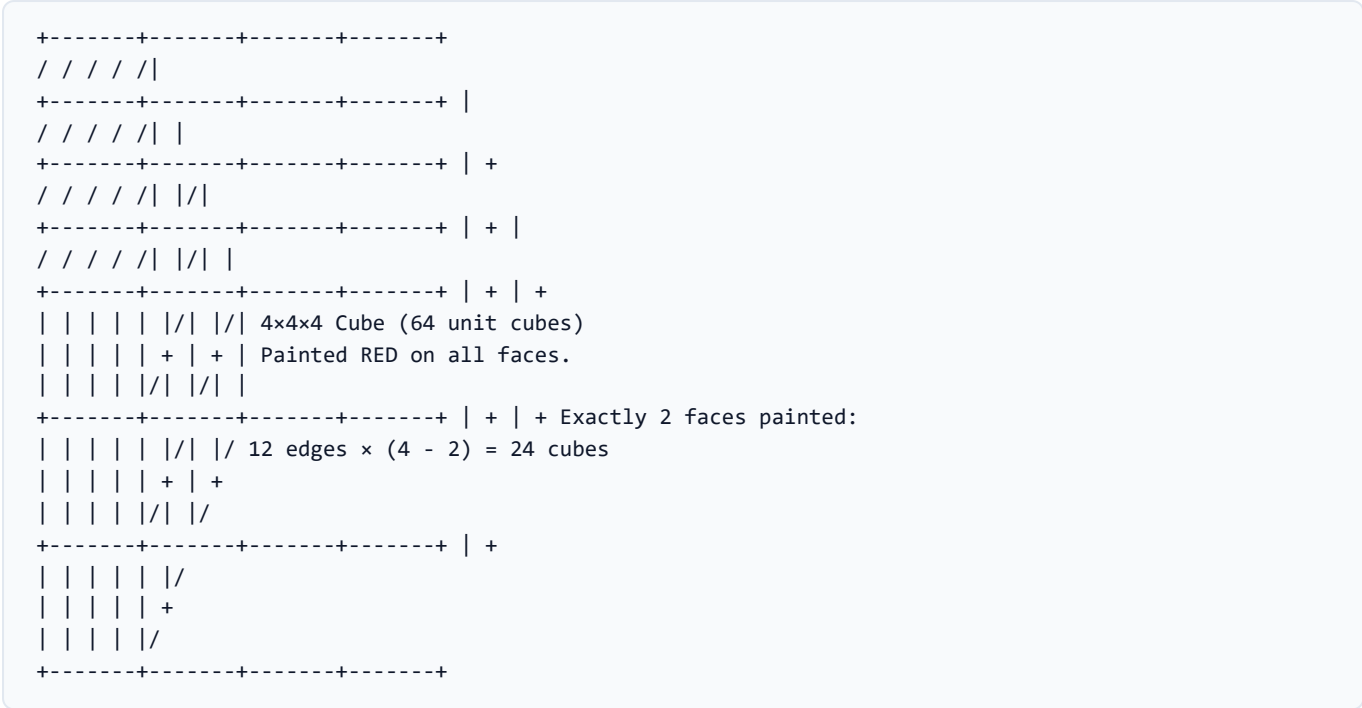
Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Options:
- **(A)** 12
- **(B)** 24
- **(C)** 36
- **(D)** 8

Correct Answer: Option **(B)** — `24`

Detailed Step-by-Step Solution & Logical Derivation:
For an n × n × n painted cube cut into unit cubes (here n = 4 since $4^3 = 64$):
Cubes with exactly two faces painted lie along the 12 edges (excluding the 2 corner cubes on each edge):
Number = 12 × (n - 2) = 12 × (4 - 2) = 12 × 2 = 24.

Question ITM-P01-Q08 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$1$_p$age$_0$2.png

Question Text:
In triangle ABC, the medians AD, BE and CF are drawn and meet at G. How many triangles are there in the figure in total?

Visual Schematic / Logical Deduction Diagram:

```
A
/ / / / | F | E
/ \ | / \ Medians AD, BE, CF meet at centroid G.
/ \ | / \ Divides ABC into 6 small triangles of
/ \ | / \ equal area:
/ \G/ \ Total distinct triangles = 16.
/_______/_|_|_______ B D C
```

Visual Schematic / Logical Deduction Diagram:


```
A
/ / / / | F | E
/ \ | / \ Medians AD, BE, CF meet at centroid G.
/ \ | / \ Divides ABC into 6 small triangles of
/ \ | / \ equal area:
/ \G/ \ Total distinct triangles = 16.
/_______/_|_|_______ B D C
```


Visual Schematic / Logical Deduction Diagram:


```
A
/ / / / | F | E
/ \ | / \ Medians AD, BE, CF meet at centroid G.
/ \ | / \ Divides ABC into 6 small triangles of
/ \ | / \ equal area:
/ \G/ \ Total distinct triangles = 16.
/_______/_|_|_______ B D C
```


Visual Schematic / Logical Deduction Diagram:


```
A
/ / / / | F | E
/ \ | / \ Medians AD, BE, CF meet at centroid G.
/ \ | / \ Divides ABC into 6 small triangles of
/ \ | / \ equal area:
/ \G/ \ Total distinct triangles = 16.
/_______/_|_|_______ B D C
```


Options:
- **(A)** 12
- **(B)** 14
- **(C)** 16
- **(D)** 18

Correct Answer: Option **(C)** — `16`

Detailed Step-by-Step Solution & Logical Derivation:
Standard median geometry counting:
In triangle ABC with medians AD, BE, CF intersecting at centroid G:
1) 6 small elementary triangles formed around centroid G: AFG, BFG, BDG, CDG, CEG, AEG.
2) 3 triangles formed by pairing two adjacent elementary triangles opposite vertices: ABG, BCG, CAG.
3) 6 triangles formed by combining three elementary triangles (split by each median into halves): ABD, ACD, BCE, ABE, CAF, CBF.

4) 1 whole outer triangle: ABC.
Total = 6 + 3 + 6 + 1 = 16 triangles.

Question ITM-P01-Q09 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
Pointing to a man, Meera said, "His mother is the only daughter of my mother." How is Meera related to the man?

Visual Schematic / Logical Deduction Diagram:

```
[Grandmother]
|
[Mother (Meera)]
|
[Son (Man)]
"His mother is the only daughter of my mother."
=> Meera is the mother of the man.
```

Visual Schematic / Logical Deduction Diagram:

```
[Grandmother]
|
[Mother (Meera)]
|
[Son (Man)]
"His mother is the only daughter of my mother."
=> Meera is the mother of the man.
```

Visual Schematic / Logical Deduction Diagram:

```
[Grandmother]
|
[Mother (Meera)]
|
[Son (Man)]
"His mother is the only daughter of my mother."
=> Meera is the mother of the man.
```

Visual Schematic / Logical Deduction Diagram:

```
[Grandmother]
|
[Mother (Meera)]
|
[Son (Man)]
"His mother is the only daughter of my mother."
=> Meera is the mother of the man.
```

Options:
- **(A)** Mother
- **(B)** Sister
- **(C)** Aunt
- **(D)** Grandmother

Correct Answer: Option **(A)** — `Mother`

Detailed Step-by-Step Solution & Logical Derivation:
Meera said: 'His mother is the only daughter of my mother.'
The 'only daughter of my mother' when spoken by a woman (Meera) refers to Meera herself.
Therefore, 'His mother is Meera.'
Thus, Meera is the man's Mother.

Question ITM-P01-Q10 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
If A + B means A is the mother of B, A − B means A is the brother of B, and A × B means A is the father of B, which expression shows that P is the maternal uncle of R?

Options:
- **(A)** P + Q − R
- **(B)** P − Q + R
- **(C)** P × Q − R
- **(D)** P − Q × R

Correct Answer: Option **(B)** — `P - Q + R`

Detailed Step-by-Step Solution & Logical Derivation:
P must be the maternal uncle of R (brother of R's mother).
- Q + R means Q is the mother of R.
- P - Q means P is the brother of Q.
Combining them: P - Q + R means P is the brother of Q (who is the mother of R), so P is R's maternal uncle.

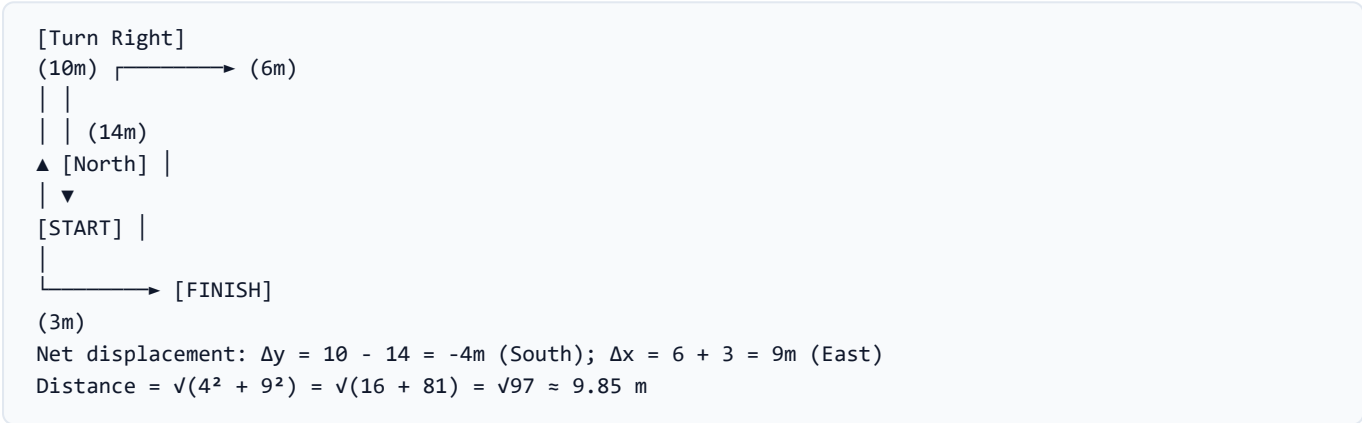
Question ITM-P01-Q11 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

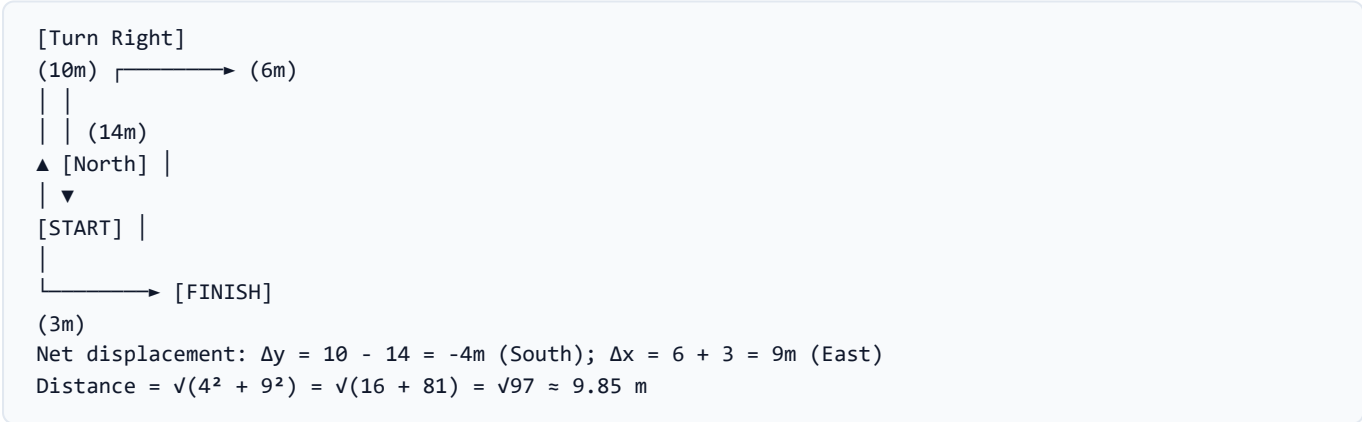
(Infosys System Engineer ITM Mock Paper 1 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
Ravi walks 10 m north, turns right and walks 6 m, turns right and walks 14 m, turns left and walks 3 m, then turns left and walks 4 m. Where is he now relative to the starting point?

Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:

```
[Turn Right]
(10m) ┌──────────► (6m)
│ │
│ │ (14m)
▲ [North] │
│ ▼
[START] │
│
└──────────► [FINISH]
(3m)
Net displacement: Δy = 10 - 14 = -4m (South); Δx = 6 + 3 = 9m (East)
Distance = √(4² + 9²) = √(16 + 81) = √97 ≈ 9.85 m
```

Visual Schematic / Logical Deduction Diagram:


```
[Turn Right]
(10m) ┌──────────► (6m)
│ │
│ │ (14m)
▲ [North] │
│ ▼
[START] │
│
└──────────► [FINISH]
(3m)
Net displacement: Δy = 10 - 14 = -4m (South); Δx = 6 + 3 = 9m (East)
Distance = √(4² + 9²) = √(16 + 81) = √97 ≈ 9.85 m
```


Options:
- **(A)** 9 m East
- **(B)** 9 m West
- **(C)** 11 m South-East
- **(D)** 5 m North

Correct Answer: Option **(A)** — `9 m East`

Detailed Step-by-Step Solution & Logical Derivation:
Step-by-step vector tracking from origin (0, 0):
1) Walks 10 m North -> (0, 10)
2) Turns right (East) and walks 6 m -> (6, 10)
3) Turns right (South) and walks 14 m -> (6, 10 - 14) = (6, -4)
4) Turns left (East) and walks 3 m -> (6 + 3, -4) = (9, -4)
5) Turns left (North) and walks 4 m -> (9, -4 + 4) = (9, 0)
Final position is (9, 0), which is exactly 9 m East of the starting point.

Question ITM-P01-Q12 `🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** `🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2

Visual Reference Plate: paper_01_page_02.png

Question Text:
A person facing North-East turns 90° clockwise, then 135° anticlockwise, then 180° clockwise. Which direction is he facing now?

Visual Schematic / Logical Deduction Diagram:

```
N (000°)
▲
NW │ NE (045°) [START]
\ │ /
\ │ /
W ──────┼──────┼──────┼──────► E (090°)
(270°) \ │ /
\ │ /
SW \ │ / SE
▼
S (180°)
Net angle = +90° - 135° + 180° = +135° clockwise.
From NE (45°) + 135° = 180° = South.
```

Visual Schematic / Logical Deduction Diagram:

```
N (000°)
▲
NW │ NE (045°) [START]
\ │ /
\ │ /
W ──────┼──────┼──────┼──────► E (090°)
(270°) \ │ /
\ │ /
SW \ │ / SE
▼
S (180°)
Net angle = +90° - 135° + 180° = +135° clockwise.
From NE (45°) + 135° = 180° = South.
```

Visual Schematic / Logical Deduction Diagram:

```
N (000°)
▲
NW │ NE (045°) [START]
\ │ /
\ │ /
W ──────┼──────┼──────┼──────► E (090°)
(270°) \ │ /
\ │ /
SW \ │ / SE
▼
S (180°)
Net angle = +90° - 135° + 180° = +135° clockwise.
From NE (45°) + 135° = 180° = South.
```

Visual Schematic / Logical Deduction Diagram:

```
N (000°)
▲
NW │ NE (045°) [START]
\ │ /
\ │ /
W ──────┼──────┼──────┼──────► E (090°)
(270°) \ │ /
\ │ /
SW \ │ / SE
▼
S (180°)
Net angle = +90° - 135° + 180° = +135° clockwise.
From NE (45°) + 135° = 180° = South.
```

Options:
- **(A)** North
- **(B)** South
- **(C)** South-West
- **(D)** East

Correct Answer: Option **(B)** — `South`

Detailed Step-by-Step Solution & Logical Derivation:
Initial direction: North-East (45° clockwise from North).
1) +90° clockwise: 45° + 90° = 135° (South-East).
2) -135° anticlockwise: 135° - 135° = 0° (North).
3) +180° clockwise: 0° + 180° = 180° (South).
Final direction is South.

Question ITM-P01-Q13 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$1$_p$age$_0$2.png

Question Text:
Find the next term: 3, 7, 16, 35, 74, ?

Options:
- **(A)** 143
- **(B)** 153
- **(C)** 149
- **(D)** 158

Correct Answer: Option **(B)** — `153`

Detailed Step-by-Step Solution & Logical Derivation:
Series: 3, 7, 16, 35, 74, ?

Pattern: ×2 + k where k increments by 1 at each step:
- 3 × 2 + 1 = 7
- 7 × 2 + 2 = 16
- 16 × 2 + 3 = 35
- 35 × 2 + 4 = 74
- Next term = 74 × 2 + 5 = 148 + 5 = 153.

Question ITM-P01-Q14 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_1_page_2.png

Question Text:
Find the next term: 4, 9, 25, 49, 121, 169, ?

Options:
- **(A)** 289
- **(B)** 259
- **(C)** 361
- **(D)** 323

Correct Answer: Option **(A)** — 289

Detailed Step-by-Step Solution & Logical Derivation:
Series: 4, 9, 25, 49, 121, 169, ?
Notice the base numbers:
- $2^2 = 4$
- $3^2 = 9$
- $5^2 = 25$
- $7^2 = 49$
- $11^2 = 121$
- $13^2 = 169$

These are the squares of consecutive prime numbers (2, 3, 5, 7, 11, 13). The next prime number is 17.
Next term = $17^2 = 289$.

Question ITM-P01-Q15 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 2

Visual Reference Plate: paper_01_page_02.png

Question Text:
Find the next letter: B, E, I, N, T, ?

Options:
- **(A)** A
- **(B)** Z
- **(C)** B
- **(D)** Y

Correct Answer: Option **(A)** — `A`

Detailed Step-by-Step Solution & Logical Derivation:
Letter positions in alphabet: B(2), E(5), I(9), N(14), T(20), ?
Differences:
- 5 - 2 = +3
- 9 - 5 = +4
- 14 - 9 = +5
- 20 - 14 = +6
Next difference is +7: 20 + 7 = 27.
In circular alphabet (mod 26): 27 ≡ 1 = A.

Question ITM-P02-Q01 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_02_page_01.png

Question Text:
Statements: All servers are machines. Some machines are laptops. No laptop is portable.
Conclusions: I. Some machines are not portable. II. Some servers are laptops. III. No server is portable.

Options:
- **(A)** Only I follows
- **(B)** I and III follow
- **(C)** I and II follow
- **(D)** None follows

Correct Answer: Option **(A)** — `Only I follows`

Detailed Step-by-Step Solution & Logical Derivation:
Some machines are laptops, and no laptop is portable, so those machines are not portable — I follows.
Nothing links servers specifically to laptops or to portability, so II and III don't follow.

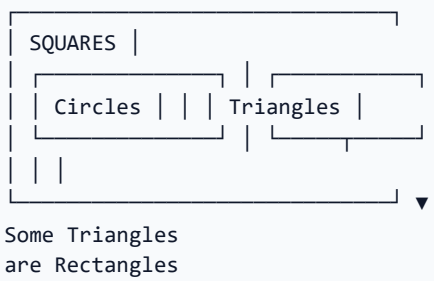
Question ITM-P02-Q02 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

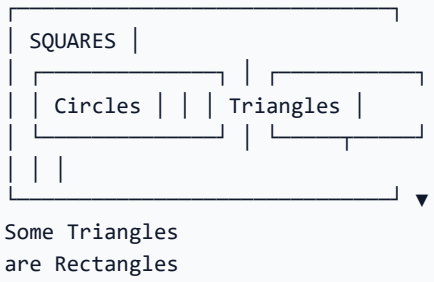
(Infosys System Engineer ITM Mock Paper 2 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_02_page_01.png

Question Text:
Statements: All circles are squares. No square is a triangle. Some triangles are rectangles.
Conclusions: I. No circle is a triangle. II. Some rectangles are not squares. III. Some squares are not triangles.

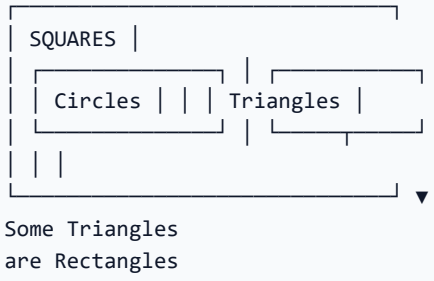
Visual Schematic / Logical Deduction Diagram:



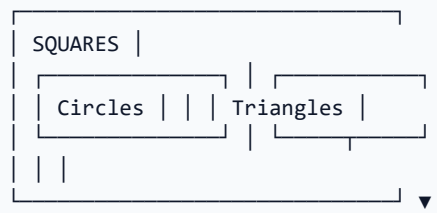
Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Options:
- **(A)** I and III only
- **(B)** I only
- **(C)** I, II and III
- **(D)** II and III only

Correct Answer: Option **(C)** — `I, II and III`

Detailed Step-by-Step Solution & Logical Derivation:
Circles
squares, and squares have no overlap with triangles, so circles have none either (I). Some
⊆
triangles are rectangles, and since no triangle is a square, those rectangles are not squares (II). Since no square is a triangle, it's also true that some squares are not triangles (III).

Question ITM-P02-Q03 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$2$_p$age$_0$1.png

Question Text:
Data Sufficiency. Is the integer n divisible by 6?
I. n is divisible by 4. II. n is divisible by 9.

Options:
- **(A)** I alone is sufficient
- **(B)** II alone is sufficient
- **(C)** Both together are sufficient, neither alone
- **(D)** Even both
together are not sufficient

Correct Answer: Option **(C)** — `Both together are sufficient, neither alone`

Detailed Step-by-Step Solution & Logical Derivation:
I alone fails (n = 4 is not divisible by 6). II alone fails (n = 9 is not divisible by 6). Together, a number divisible by both 4 and 9 is divisible by their LCM, 36, which is always divisible by 6.

Question ITM-P02-Q04 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$2$_p$age$_0$1.png

Question Text:
Data Sufficiency. What is the cost price of an article?
I. It was sold at a profit of 20%. II. If sold at ₹150 more, the profit would have been 32%.

Options:
- **(A)** I alone is sufficient
- **(B)** II alone is sufficient
- **(C)** Both together are sufficient, neither alone
- **(D)** Even both
together are not sufficient

Correct Answer: Option **(B)** — `II alone is sufficient`

Detailed Step-by-Step Solution & Logical Derivation:
II alone: 12% of CP = ₹150, so CP = ₹1,250 — sufficient. I alone gives only a percentage, not an absolute value.

Question ITM-P02-Q05 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$2$_p$age$_0$1.png

Question Text:
In a code, every letter is shifted 3 places forward in the alphabet (wrapping Z→C), so FLOWER is written as IORZHU. Using the same code, how is GARDEN written?

Options:
- **(A)** JDUGHQ
- **(B)** JDUFHQ
- **(C)** JCUGHQ
- **(D)** JDVGHQ

Correct Answer: Option **(A)** — `JDUGHQ`

Detailed Step-by-Step Solution & Logical Derivation:
Applying the same +3 shift: G→J, A→D, R→U, D→G, E→H, N→Q, giving JDUGHQ.

Question ITM-P02-Q06 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$2$_p$age$_0$1.png

Question Text:
In a code, "data flows fast" = "zu ka mo", "fast network grows" = "ka ta ri", "network data secure" = "zu ta ve". What is the code for "grows secure data"?

Options:
- **(A)** ri ve zu
- **(B)** ta ve zu
- **(C)** ri ve ka
- **(D)** ka ve zu

Correct Answer: Option **(A)** — `ri ve zu`

Detailed Step-by-Step Solution & Logical Derivation:
"fast" is common to statements 1 and 2 → ka. "data" is common to 1 and 3 → zu. "network" is common to 2 and 3 → ta. By elimination: flows = mo, grows = ri, secure = ve. So "grows secure data" = ri ve zu.

Question ITM-P02-Q07 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$2$_p$age$_0$1.png

Question Text:
A cube is painted red on one pair of opposite faces, blue on another pair, and green on the last pair. It is then cut into 27 identical small cubes. How many small cubes have exactly one face painted?

Visual Schematic / Logical Deduction Diagram:

```
Top/Bottom: Green
Front/Back: Red
Left/Right: Blue
4×4×4 Cube (64 unit cubes)
At least two different colors = (Corners: 8) + (Edges: 24) = 32 cubes.
```

Visual Schematic / Logical Deduction Diagram:


```
Top/Bottom: Green
Front/Back: Red
Left/Right: Blue
4×4×4 Cube (64 unit cubes)
At least two different colors = (Corners: 8) + (Edges: 24) = 32 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
Top/Bottom: Green
Front/Back: Red
Left/Right: Blue
4×4×4 Cube (64 unit cubes)
At least two different colors = (Corners: 8) + (Edges: 24) = 32 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
Top/Bottom: Green
Front/Back: Red
Left/Right: Blue
4×4×4 Cube (64 unit cubes)
At least two different colors = (Corners: 8) + (Edges: 24) = 32 cubes.
```


Options:
- **(A)** 6
- **(B)** 8
- **(C)** 9
- **(D)** 12

Correct Answer: Option **(A)** — `6`

Detailed Step-by-Step Solution & Logical Derivation:
In a 3×3×3 cube, only the centre cube of each face touches just one painted face — 1 per face × 6 faces = 6, regardless of the colour scheme.

Question ITM-P02-Q08 `🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** `🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$2$_p$age$_0$1.png

Question Text:
A rectangle is divided into a 3 × 3 grid of smaller cells by 2 evenly spaced vertical lines and 2 evenly spaced horizontal lines drawn inside it. How many rectangles (of any size) are there in the whole figure?

Options:
- **(A)** 30
- **(B)** 36
- **(C)** 27
- **(D)** 24

Correct Answer: Option **(B)** — `36`

Detailed Step-by-Step Solution & Logical Derivation:
With 2 interior + 2 boundary lines in each direction, there are 4 vertical and 4 horizontal lines.
Rectangles = C(4,2) × C(4,2) = 6 × 6 = 36.

Question ITM-P02-Q09 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$2$_p$age$_0$2.png

Question Text:
Pointing to a photograph, a man says, "She is the daughter of my grandfather's only son." How is the woman in the photograph related to the man?

Visual Schematic / Logical Deduction Diagram:

```
[Grandfather]
|
[Only Son] (Father)
|
┌─────────┴─────────┐
| |
[Man] [Daughter] (Sister)
=> "She is the daughter of my grandfather's only son" = Sister.
```

Visual Schematic / Logical Deduction Diagram:

```
[Grandfather]
|
[Only Son] (Father)
|
┌─────────┴─────────┐
| |
[Man] [Daughter] (Sister)
=> "She is the daughter of my grandfather's only son" = Sister.
```

Visual Schematic / Logical Deduction Diagram:

```
[Grandfather]
|
[Only Son] (Father)
|
┌──────────┴──────────┐
| |
[Man] [Daughter] (Sister)
=> "She is the daughter of my grandfather's only son" = Sister.
```

Visual Schematic / Logical Deduction Diagram:


```
[Grandfather]
|
[Only Son] (Father)
|
┌──────────┴──────────┐
| |
[Man] [Daughter] (Sister)
=> "She is the daughter of my grandfather's only son" = Sister.
```


Options:
- **(A)** Sister
- **(B)** Daughter
- **(C)** Niece
- **(D)** Mother

Correct Answer: Option **(A)** — `Sister`

Detailed Step-by-Step Solution & Logical Derivation:
The grandfather's only son is the man's father. The daughter of the man's father is his sister.

Question ITM-P02-Q10 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$2$_p$age$_0$2.png

Question Text:
If A ÷ B means A is the daughter of B, A + B means A is the son of B, and A − B means A is the wife of B, which expression correctly shows that M is the granddaughter of N?

Options:
- **(A)** M ÷ P + N
- **(B)** M + P ÷ N
- **(C)** M − P ÷ N
- **(D)** M ÷ P − N

Correct Answer: Option **(A)** — `M ÷ P + N`

Detailed Step-by-Step Solution & Logical Derivation:
M ÷ P: M is daughter of P. P + N: P is son of N. So M is P's daughter, and P is N's son — M is N's granddaughter.

Question ITM-P02-Q11 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Priya starts at a point, faces south and walks 8 m. She turns left and walks 5 m, turns left again and walks 8 m, then turns right and walks 3 m. How far is she from the starting point, and in which direction?

Visual Schematic / Logical Deduction Diagram:

```
[START]
|
▼ (8m South)
|
(5m) ←┘ [Turn Left = East]
▲
| (12m North)
|
[FINISH]
Net displacement: Δy = 12 - 8 = 4m North; Δx = 5m East
Distance = √(4² + 5²) = √(16 + 25) = √41 m
```

Visual Schematic / Logical Deduction Diagram:

```
[START]
|
▼ (8m South)
|
(5m) ←┘ [Turn Left = East]
▲
| (12m North)
|
[FINISH]
Net displacement: Δy = 12 - 8 = 4m North; Δx = 5m East
Distance = √(4² + 5²) = √(16 + 25) = √41 m
```

Visual Schematic / Logical Deduction Diagram:

```
[START]
|
▼ (8m South)
|
(5m) ◄──┘ [Turn Left = East]
▲
| (12m North)
|
[FINISH]
Net displacement: Δy = 12 - 8 = 4m North; Δx = 5m East
Distance = √(4² + 5²) = √(16 + 25) = √41 m
```

Visual Schematic / Logical Deduction Diagram:


```
[START]
|
▼ (8m South)
|
(5m) ◄──┘ [Turn Left = East]
▲
| (12m North)
|
[FINISH]
Net displacement: Δy = 12 - 8 = 4m North; Δx = 5m East
Distance = √(4² + 5²) = √(16 + 25) = √41 m
```


Options:
- **(A)** 8 m East
- **(B)** 8 m West
- **(C)** 2 m East
- **(D)** 2 m West

Correct Answer: Option **(A)** — `8 m East`

Detailed Step-by-Step Solution & Logical Derivation:
Tracking positions from (0,0): south 8 → (0,−8); left turn faces east, walk 5 → (5,−8); left turn faces north, walk 8 → (5,0); right turn faces east, walk 3 → (8,0). Net displacement is 8 m East.

Question ITM-P02-Q12 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 2 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A man facing South turns 45° anticlockwise, then 90° clockwise, then 225° anticlockwise. Which direction is he facing now?

Options:
- **(A)** North
- **(B)** South-East
- **(C)** South-West
- **(D)** West

Correct Answer: Option **(A)** — `North`

Detailed Step-by-Step Solution & Logical Derivation:
Using bearings (clockwise from North): South = 180°. −45° (anticlockwise) = 135°; +90° (clockwise) = 225°; −225° (anticlockwise) = 0°, which is North.

Question ITM-P02-Q13 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 2 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Find the next term: 2, 6, 15, 31, 56, ?

Options:
- **(A)** 92
- **(B)** 88
- **(C)** 90
- **(D)** 84

Correct Answer: Option **(A)** — `92`

Detailed Step-by-Step Solution & Logical Derivation:
Differences are 4, 9, 16, 25 (perfect squares), so the next difference is 36: 56 + 36 = 92.

Question ITM-P02-Q14 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 2 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Find the next term: 5, 11, 23, 47, 95, ?

Options:
- **(A)** 189
- **(B)** 191
- **(C)** 193
- **(D)** 190

Correct Answer: Option **(B)** — `191`

Detailed Step-by-Step Solution & Logical Derivation:
Each term = previous × 2 + 1: 95 × 2 + 1 = 191.

Question ITM-P02-Q15 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 2 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$2$_p$age$_0$2.png

Question Text:
Find the next letter-pair: CE, FI, IM, LQ, ?

Options:
- **(A)** OT
- **(B)** OU
- **(C)** NT
- **(D)** PU

Correct Answer: Option **(B)** — `OU`

Detailed Step-by-Step Solution & Logical Derivation:
First letters C, F, I, L rise by 3 each time → next is O. Second letters E, I, M, Q rise by 4 each time → next is U. Answer: OU.

Question ITM-P03-Q01 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$3$_p$age$_0$1.png

Question Text:
Statements: All P are Q. All Q are R. Some R are S. No S is T.
Conclusions: I. All P are R. II. Some Q are S. III. No P is T. IV. Some R are not T.

Options:
- **(A)** Only I follows
- **(B)** I and IV follow
- **(C)** I, II and IV follow
- **(D)** All follow

Correct Answer: Option **(B)** — `I and IV follow`

Detailed Step-by-Step Solution & Logical Derivation:
Since P⊆Q⊆R, "All P are R" (I) follows directly. Some R are S, and no S is T, so those particular R-elements are not T — "Some R are not T" (IV) follows. Neither II nor III can be derived, since we don't know how P or Q specifically relates to the R-elements that happen to be S.

Question ITM-P03-Q02 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$3$_p$age$_0$1.png

Question Text:
Statements: All coins are metals. No metal is wood. Some woods are furniture.
Conclusions: I. No coin is wood. II. Some furniture is not metal.

Options:
- **(A)** Only I follows
- **(B)** Only II follows
- **(C)** Both I and II follow
- **(D)** Neither follows

Correct Answer: Option **(C)** — `Both I and II follow`

Detailed Step-by-Step Solution & Logical Derivation:
Coins⊆metals, and metals∩wood=∅, so coins∩wood=∅ — I follows. Some woods are furniture, and since no metal is wood, those wood-furniture items are not metal — II follows.

Question ITM-P03-Q03 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$3$_p$age$_0$1.png

Question Text:
Data Sufficiency. What is the two-digit number N?
I. The sum of its digits is 9.
II. The number obtained by reversing its digits is 27 less than N.

Options:
- **(A)** Statement I alone is sufficient, but II alone is not
- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Each statement alone is sufficient
(e) Even both statements together are not sufficient

Correct Answer: Option **(C)** — `Both statements together are sufficient, but neither alone is`

Detailed Step-by-Step Solution & Logical Derivation:
II alone: N − reversed = 9(tens digit − units digit) = 27, so tens − units = 3 — multiple numbers fit (41, 52, 63, 74, 85, 96), insufficient alone. I alone: digit sum = 9 also fits multiple numbers, insufficient alone. Together: tens − units = 3 and tens + units = 9 gives tens = 6, units = 3, so N = 63 — unique.

Question ITM-P03-Q04 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$3$_p$age$_0$1.png

Question Text:
Data Sufficiency. Is the integer x even?
I. x/2 is an integer.
II. x^2 is even.

Options:
- **(A)** Statement I alone is sufficient, but II alone is not
- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Each statement alone is sufficient
(e) Even both statements together are not sufficient

Correct Answer: Option **(D)** — `Each statement alone is sufficient (e) Even both statements together are not sufficient`

Detailed Step-by-Step Solution & Logical Derivation:
I alone: x/2 being an integer directly means x is even — sufficient. II alone: x^2 even is only possible if x itself is even (an odd number squared is always odd) — sufficient. Each statement alone is sufficient.

Question ITM-P03-Q05 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$3$_p$age$_0$1.png

Question Text:
In a certain code, TRAIN is written as 23145, GAME is written as 6178, and MEAT is written as 7812.
What is the code for GATE?

Options:
- **(A)** 6128
- **(B)** 6218
- **(C)** 6182
- **(D)** 8126

Correct Answer: Option **(A)** — `6128`

Detailed Step-by-Step Solution & Logical Derivation:
From TRAIN=23145: T=2,R=3,A=1,I=4,N=5. From GAME=6178: G=6,A=1,M=7,E=8. From MEAT=7812: confirms M=7,E=8,A=1,T=2. So GATE = G(6)A(1)T(2)E(8) = 6128.

Question ITM-P03-Q06 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$3$_p$age$_0$2.png

Question Text:
In a code: "bright light shines" = "mo ta pu", "shines through night" = "pu ri ke", "night bright stars" = "ke mo lo", "stars through galaxy" = "lo ri zu". What is the code for "night light galaxy"?

Options:
- **(A)** ke ta zu
- **(B)** ke ta ri
- **(C)** mo ta zu
- **(D)** ke lo zu

Correct Answer: Option **(A)** — `ke ta zu`

Detailed Step-by-Step Solution & Logical Derivation:
Comparing statement 1 & 2, the common word "shines" gets the common code "pu". Comparing 2 &

3, "night" gets the common code "ke". Comparing 1 & 3, "bright" gets the common code "mo". This leaves light=ta, stars=lo, through=ri, galaxy=zu (verified consistent with statement 4). So "night light galaxy" = ke ta zu.

Question ITM-P03-Q07 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
A cuboid measuring 5 × 4 × 3 units is painted on all its outer surfaces and then cut into unit cubes. How many unit cubes have exactly 2 painted faces?

Visual Schematic / Logical Deduction Diagram:

```
5 × 4 × 3 Cuboid (60 unit cubes)
Total Painted Surface Cubes:
No face painted = (5-2) × (4-2) × (3-2) = 3 × 2 × 1 = 6 cubes.
Painted cubes = 60 - 6 = 54 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
5 × 4 × 3 Cuboid (60 unit cubes)
Total Painted Surface Cubes:
No face painted = (5-2) × (4-2) × (3-2) = 3 × 2 × 1 = 6 cubes.
Painted cubes = 60 - 6 = 54 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
5 × 4 × 3 Cuboid (60 unit cubes)
Total Painted Surface Cubes:
No face painted = (5-2) × (4-2) × (3-2) = 3 × 2 × 1 = 6 cubes.
Painted cubes = 60 - 6 = 54 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
5 × 4 × 3 Cuboid (60 unit cubes)
Total Painted Surface Cubes:
No face painted = (5-2) × (4-2) × (3-2) = 3 × 2 × 1 = 6 cubes.
Painted cubes = 60 - 6 = 54 cubes.
```

Options:
- **(A)** 24
- **(B)** 22
- **(C)** 20
- **(D)** 26

Correct Answer: Option **(A)** — `24`

Detailed Step-by-Step Solution & Logical Derivation:
For a cuboid a×b×c, the number of unit cubes with exactly 2 painted faces = 4[(a−2)+(b−2)+(c−2)]. With a=5,b=4,c=3: 4[3+2+1] = 4×6 = 24.

Question ITM-P03-Q08 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$3$_p$age$_0$2.png

Question Text:
How many triangles are there in total in a complete five-pointed star (pentagram) figure?

Options:
- **(A)** 5
- **(B)** 10
- **(C)** 35
- **(D)** 15

Correct Answer: Option **(C)** — `35`

Detailed Step-by-Step Solution & Logical Derivation:
A complete five-pointed star (pentagram) figure contains 35 triangles in total (5 small tip triangles, 5 medium, 5 larger, and combinations up to the whole star) — a standard, well-documented result for this figure.

Question ITM-P03-Q09 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$3$_p$age$_0$2.png

Question Text:
Introducing a man, a woman says, "His wife is the only daughter-in-law of my father." How is the woman related to the man's wife?

Options:
- **(A)** Sister-in-law
- **(B)** Sister

- **(C)** Mother
- **(D)** Aunt

Correct Answer: Option **(A)** — `Sister-in-law`

Detailed Step-by-Step Solution & Logical Derivation:
The man's wife is the only daughter-in-law of the woman's father, meaning the woman's father has only one son — the man himself. So the man is the woman's brother, making the woman the sister-in-law of the man's wife.

Question ITM-P03-Q10 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$3$_p$age$_0$2.png

Question Text:
If A % B means A is the mother of B, and A * B means A is the father of B, which expression shows that C is the paternal grandmother of D?

Options:
- **(A)** C % E * D
- **(B)** C * E % D
- **(C)** E % C * D
- **(D)** D % E * C

Correct Answer: Option **(A)** — `C % E * D`

Detailed Step-by-Step Solution & Logical Derivation:
C % E: C is mother of E. E * D: E is father of D. So C is the mother of D's father, i.e., C is D's paternal grandmother.

Question ITM-P03-Q11 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$3$_p$age$_0$2.png

Question Text:
A man walks 12 m North, then 9 m East, then 8 m South, then 6 m West. How far is he from the starting point, and in which general direction?

Visual Schematic / Logical Deduction Diagram:

```
(12m North)
┌──────────────────► (9m East)
│ │
▲ ▼ (8m South)
[START] ◄──────────┘
(6m West)
Net displacement: Δy = 12 - 8 = 4m North; Δx = 9 - 6 = 3m East
Shortest distance = √(3² + 4²) = 5 m.
```

Visual Schematic / Logical Deduction Diagram:

```
(12m North)
┌──────────────────► (9m East)
│ │
▲ ▼ (8m South)
[START] ◄──────────┘
(6m West)
Net displacement: Δy = 12 - 8 = 4m North; Δx = 9 - 6 = 3m East
Shortest distance = √(3² + 4²) = 5 m.
```

Visual Schematic / Logical Deduction Diagram:

```
(12m North)
┌──────────────────► (9m East)
│ │
▲ ▼ (8m South)
[START] ◄──────────┘
(6m West)
Net displacement: Δy = 12 - 8 = 4m North; Δx = 9 - 6 = 3m East
Shortest distance = √(3² + 4²) = 5 m.
```

Visual Schematic / Logical Deduction Diagram:

```
(12m North)
┌──────────────────► (9m East)
│ │
▲ ▼ (8m South)
[START] ◄──────────┘
(6m West)
Net displacement: Δy = 12 - 8 = 4m North; Δx = 9 - 6 = 3m East
Shortest distance = √(3² + 4²) = 5 m.
```

Options:

- **(A)** 5 m, North-East
- **(B)** 5 m, North-West
- **(C)** 7 m, North-East
- **(D)** 5 m, South-East

Correct Answer: Option **(A)** — `5 m, North-East`

Detailed Step-by-Step Solution & Logical Derivation:

Net North-South displacement = 12−8 = 4 m North; net East-West = 9−6 = 3 m East. Distance = $\sqrt{(3^2+4^2)} = \sqrt{25} = 5$ m, in the general North-East direction.

Question ITM-P03-Q12 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$3$_p$age$_0$2.png

Question Text:
A cyclist starts facing East. He turns 135° clockwise, then 90° anticlockwise, then 180° clockwise. Which direction is he facing now?

Options:
- **(A)** North-West
- **(B)** South-East
- **(C)** South-West
- **(D)** North-East

Correct Answer: Option **(A)** — `North-West`

Detailed Step-by-Step Solution & Logical Derivation:
Using bearings (clockwise from North, East=90°): 90°+135°=225°(SW); 225°−90°=135°(SE); 135°+180°=315°, which is North-West.

Question ITM-P03-Q13 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$3$_p$age$_0$2.png

Question Text:
Find the next term: 7, 26, 63, 124, 215, ?

Options:
- **(A)** 342
- **(B)** 344
- **(C)** 341
- **(D)** 343

Correct Answer: Option **(A)** — `342`

Detailed Step-by-Step Solution & Logical Derivation:
Each term is n^3-1 for n=2,3,4,5,6: 7,26,63,124,215; next (n=7): 343−1 = 342.

Question ITM-P03-Q14 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$3$_p$age$_0$2.png

Question Text:
Find the next term: 4, 5, 9, 18, 34, ?

Options:
- **(A)** 55
- **(B)** 59
- **(C)** 63
- **(D)** 57

Correct Answer: Option **(B)** — `59`

Detailed Step-by-Step Solution & Logical Derivation:
Differences are 1,4,9,16 (perfect squares); next difference is 25: 34+25 = 59.

Question ITM-P03-Q15 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$3$_p$age$_0$2.png

Question Text:
Find the next term: B, D, G, K, P, ?

Options:
- **(A)** V
- **(B)** U
- **(C)** T
- **(D)** W

Correct Answer: Option **(A)** — `V`

Detailed Step-by-Step Solution & Logical Derivation:
Differences between consecutive letters (by position) increase by 1 each time: +2,+3,+4,+5, next +6:
P(16)+6 = 22 = V.

Question ITM-P04-Q01 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$4$_p$age$_0$1.png

Question Text:
Statements: Some pens are pencils. All pencils are erasers. No eraser is a sharpener.
Conclusions: I. No pencil is a sharpener. II. Some pens are erasers. III. Some erasers are not pens. IV. Either some pens are sharpeners or no pens are sharpeners.

Options:
- **(A)** Only I and II follow
- **(B)** I, II and IV follow
- **(C)** All follow
- **(D)** Only IV follows

Correct Answer: Option **(B)** — `I, II and IV follow`

Detailed Step-by-Step Solution & Logical Derivation:
Pencils⊆erasers and erasers∩sharpeners=∅, so pencils∩sharpeners=∅ (I follows). Some pens are pencils, and pencils⊆erasers, so some pens are erasers (II follows). III isn't guaranteed (no information ensures an eraser outside the pen-pencil overlap exists). For IV: we know the pen-pencils aren't sharpeners, but nothing is said about pens that aren't pencils, so neither "some pens are sharpeners" nor "no pens are sharpeners" can be individually proven — exactly the condition for the standard either-or rule, so IV follows.

Question ITM-P04-Q02 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$4$_p$age$_0$1.png

Question Text:
Statements: Only members are allowed entry. Some members are officers. All officers are trained.
Conclusions: I. Some trained persons are members. II. All those allowed entry are trained. III. Some members are trained.

Options:
- **(A)** I and III only
- **(B)** II and III only
- **(C)** All follow
- **(D)** None follows

Correct Answer: Option **(A)** — `I and III only`

Detailed Step-by-Step Solution & Logical Derivation:
"Only members are allowed entry" means allowed-entry⊆members. Some members are officers, and officers⊆trained, so those members are trained — this directly gives "some members are trained" (III) and, by simple conversion, "some trained are members" (I). But since only some members are officers/trained, we cannot say all those allowed entry are trained (II fails, as non-officer members may not be trained).

Question ITM-P04-Q03 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$4$_p$age$_0$1.png

Question Text:
Data Sufficiency. What is the value of x?
I. $x^2 - 5x + 6 = 0$
II. $x > 2$

Options:
- **(A)** Statement I alone is sufficient, but II alone is not
- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Even both statements together are not sufficient

Correct Answer: Option **(C)** — `Both statements together are sufficient, but neither alone is`

Detailed Step-by-Step Solution & Logical Derivation:
I alone: $x^2-5x+6=0$ factors to $(x-2)(x-3)=0$, giving $x=2$ or $x=3$ — two solutions, insufficient. II alone: $x>2$ alone is insufficient. Together: only $x=3$ satisfies both $x>2$ and the equation — sufficient together.

Question ITM-P04-Q04 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$4$_p$age$_0$1.png

Question Text:
Data Sufficiency. In triangle ABC, what is the measure of angle B?
I. Angle A = 50°, and the triangle is isosceles with AB = AC.
II. Angle C = 65°.

Options:
- **(A)** Statement I alone is sufficient, but II alone is not
- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Even both statements together are not sufficient

Correct Answer: Option **(A)** — `Statement I alone is sufficient, but II alone is not`

Detailed Step-by-Step Solution & Logical Derivation:
I alone: AB=AC means the angles opposite them are equal, so angle B = angle C; with angle A=50°, B+C=130°, and B=C gives B=65° — sufficient. II alone gives only one angle with no other relation — insufficient.

Question ITM-P04-Q05 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$4$_p$age$_0$2.png

Question Text:
In a code, each letter is first replaced by its position in the alphabet, and that number is then doubled and reduced by 1. Using this rule, CAT is coded as 5-1-39. What is the code for DOG?

Options:
- **(A)** 7-29-13
- **(B)** 8-29-13
- **(C)** 7-30-13
- **(D)** 7-29-14

Correct Answer: Option **(A)** — `7-29-13`

Detailed Step-by-Step Solution & Logical Derivation:
D→4→2(4)−1=7. O→15→2(15)−1=29. G→7→2(7)−1=13. Code: 7-29-13.

Question ITM-P04-Q06 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$4$_p$age$_0$2.png

Question Text:
In a code: "alpha beta gamma delta" = "zx qr mn kl", "beta gamma epsilon" = "qr mn vw", "gamma delta zeta" = "mn kl tu", "epsilon zeta eta" = "vw tu yz", "alpha eta theta" = "zx yz op". What is the code for "theta beta zeta"?

Options:
- **(A)** op qr tu
- **(B)** op qr vw
- **(C)** zx qr tu
- **(D)** op mn tu

Correct Answer: Option **(A)** — `op qr tu`

Detailed Step-by-Step Solution & Logical Derivation:
Comparing statements pairwise for common words and codes: shines=pu, gamma=mn, bright=mo (from statements 1&3), giving delta=kl (from statements 1&3's remaining elements), light=ta, zeta=tu, epsilon=vw, eta=yz, alpha=zx, theta=op. So "theta beta zeta" = op qr tu.

Question ITM-P04-Q07 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
A cube of side 6 units is painted red on all faces and then cut into 1-unit cubes. How many of the small cubes have at least 2 painted faces?

Visual Schematic / Logical Deduction Diagram:

```
6 × 6 × 6 Cube (216 unit cubes)
Painted RED on all faces.
Cubes with NO paint = (6 - 2)³ = 4³ = 64 cubes.
Cubes with at least 1 face painted = 216 - 64 = 152 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
6 × 6 × 6 Cube (216 unit cubes)
Painted RED on all faces.
Cubes with NO paint = (6 - 2)³ = 4³ = 64 cubes.
Cubes with at least 1 face painted = 216 - 64 = 152 cubes.
```

Visual Schematic / Logical Deduction Diagram:

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Visual Schematic / Logical Deduction Diagram:


```
6 × 6 × 6 Cube (216 unit cubes)
Painted RED on all faces.
Cubes with NO paint = (6 - 2)³ = 4³ = 64 cubes.
Cubes with at least 1 face painted = 216 - 64 = 152 cubes.
```


Options:
- **(A)** 56
- **(B)** 48
- **(C)** 64
- **(D)** 60

Correct Answer: Option **(A)** — `56`

Detailed Step-by-Step Solution & Logical Derivation:
For an n×n×n cube, cubes with exactly 2 painted faces (edges) = 12(n−2); with exactly 3 (corners) = 8.
For n=6: edges=12×4=48, corners=8, so at least 2 = 48+8 = 56.

Question ITM-P04-Q08 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_{0}$4$_{p}$age$_{0}$2.png

Question Text:
How many squares (of all sizes) are there in a 4×4 grid of unit squares?

Options:
- **(A)** 30
- **(B)** 25
- **(C)** 36
- **(D)** 20

Correct Answer: Option **(A)** — `30`

Detailed Step-by-Step Solution & Logical Derivation:
Squares of size k in an n×n grid = $(n-k+1)^2$. For n=4: k=1→16, k=2→9, k=3→4, k=4→1. Total = 16+9+4+1 = 30.

Question ITM-P04-Q09 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
P is the father of Q. Q is the brother of R. S is the mother of R. T is the husband of S. How is P related to T?

Options:
- **(A)** Father
- **(B)** P and T are the same person
- **(C)** Brother-in-law
- **(D)** Cannot be determined

Correct Answer: Option **(B)** — `P and T are the same person`

Detailed Step-by-Step Solution & Logical Derivation:
Q and R are siblings (Q is R's brother), so S (R's mother) is also Q's mother. P is Q's father, so P and S are Q's parents — meaning P is S's husband. Since T is defined as S's husband, P and T are the same person.

Question ITM-P04-Q10 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
If A + B means A is the father of B, A − B means A is the mother of B, and A × B means A is the son of B, which expression shows that M is the maternal grandfather of N?

Options:
- **(A)** M + Q − N
- **(B)** M − Q + N
- **(C)** Q + M − N
- **(D)** M × Q − N

Correct Answer: Option **(A)** — `M + Q - N`

Detailed Step-by-Step Solution & Logical Derivation:
M+Q: M is father of Q. Q−N: Q is mother of N. So M is father of Q, who is N's mother — M is N's

maternal grandfather.

Question ITM-P04-Q11 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
A man walks 14 m East, then 20 m North, then 5 m West, then 8 m South. How far is he from the starting point, and in which direction?

Visual Schematic / Logical Deduction Diagram:

```
(20m North)
┌──────────────────────► [FINISH]
│ ▲
│ │
[START] ──┴──────────────────────┘ (14m East - 5m West = 9m)
(14m East) (8m South)
Net displacement: Δy = 20 - 8 = 12m North; Δx = 14 - 5 = 9m East
Shortest distance = √(9² + 12²) = √(81 + 144) = √225 = 15 m.
```

Visual Schematic / Logical Deduction Diagram:

```
(20m North)
┌──────────────────────► [FINISH]
│ ▲
│ │
[START] ──┴──────────────────────┘ (14m East - 5m West = 9m)
(14m East) (8m South)
Net displacement: Δy = 20 - 8 = 12m North; Δx = 14 - 5 = 9m East
Shortest distance = √(9² + 12²) = √(81 + 144) = √225 = 15 m.
```

Visual Schematic / Logical Deduction Diagram:

```
(20m North)
┌──────────────────────► [FINISH]
│ ▲
│ │
[START] ──┴──────────────────────┘ (14m East - 5m West = 9m)
(14m East) (8m South)
Net displacement: Δy = 20 - 8 = 12m North; Δx = 14 - 5 = 9m East
Shortest distance = √(9² + 12²) = √(81 + 144) = √225 = 15 m.
```

Visual Schematic / Logical Deduction Diagram:

```
(20m North)
┌──────────────────► [FINISH]
│ ▲
│ │
[START] ──┴──────────────────┘ (14m East - 5m West = 9m)
(14m East) (8m South)
Net displacement: Δy = 20 - 8 = 12m North; Δx = 14 - 5 = 9m East
Shortest distance = √(9² + 12²) = √(81 + 144) = √225 = 15 m.
```

Options:
- **(A)** 15 m, North-East
- **(B)** 15 m, North-West
- **(C)** 13 m, North-East
- **(D)** 17 m, North-East

Correct Answer: Option **(A)** — `15 m, North-East`

Detailed Step-by-Step Solution & Logical Derivation:
Net East-West = 14−5 = 9 m East; net North-South = 20−8 = 12 m North. Distance = $\sqrt{(9^2+12^2)} = \sqrt{225} =$ 15 m, direction North-East.

Question ITM-P04-Q12 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$4$_p$age$_0$2.png

Question Text:
A ship sails 10 km from port on a bearing of 030° (N30°E), then sails 10 km on a bearing of 150° (S30°E). What is the ship's final position relative to the port?

Options:
- **(A)** 10 km due East (bearing 090°)
- **(B)** 10 km due North
- **(C)** 17.3 km, bearing 060°
- **(D)** 10 km, bearing 120°

Correct Answer: Option **(A)** — `10 km due East (bearing 090°)`

Detailed Step-by-Step Solution & Logical Derivation:
Leg 1 (bearing 030°, 10 km): East = 10 sin30° = 5, North = 10 cos30° = 5√3. Leg 2 (bearing 150°, 10 km): East = 10 sin150° = 5, North = 10 cos150° = −5√3. Totals: East = 5+5 = 10, North = 5√3−5√3 = 0. Final position: 10 km due East (bearing 090°).

Question ITM-P04-Q13 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
Find the next term: 2, 5, 12, 27, 58, ?

Options:
- **(A)** 121
- **(B)** 125
- **(C)** 115
- **(D)** 118

Correct Answer: Option **(A)** — `121`

Detailed Step-by-Step Solution & Logical Derivation:
Differences are 3, 7, 15, 31 — each is 2^n-1 (n=2,3,4,5); next difference is $2^6-1=63$. 58+63 = 121.

Question ITM-P04-Q14 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
Find the next term: 1, 3, 7, 17, 41, ?

Options:
- **(A)** 99
- **(B)** 95
- **(C)** 101
- **(D)** 89

Correct Answer: Option **(A)** — `99`

Detailed Step-by-Step Solution & Logical Derivation:
Each term follows a(n) = 2×a(n−1) + a(n−2): 2×17+7=41 ✓; next = 2×41+17 = 99.

Question ITM-P04-Q15 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$4$_p$age$_0$2.png

Question Text:
Find the next term: Y, V, R, M, ?

Options:
- **(A)** G
- **(B)** H
- **(C)** F
- **(D)** I

Correct Answer: Option **(A)** — `G`

Detailed Step-by-Step Solution & Logical Derivation:
Gaps between letters increase by 1 each time: −3, −4, −5, next −6. M(13)−6 = 7 = G.

Question ITM-P05-Q01 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$5$_p$age$_0$1.png

Question Text:
Statements: All toys are games. All games are activities. Some crafts are not activities. No craft is a hobby.
Conclusions: I. All toys are activities. II. Some crafts are not hobbies. III. No toy is a craft.

Options:
- **(A)** Only I follows
- **(B)** I and II follow
- **(C)** I, II and III follow
- **(D)** None follows

Correct Answer: Option **(B)** — `I and II follow`

Detailed Step-by-Step Solution & Logical Derivation:
Toys⊆games⊆activities, so all toys are activities (I follows). "Some crafts are not activities" confirms

crafts exist, and since no craft is a hobby, those crafts are also not hobbies — "some crafts are not hobbies" follows (II). III is not guaranteed, since nothing links toys specifically to crafts.

Question ITM-P05-Q02 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$5$_p$age$_0$1.png

Question Text:
Statements: All researchers are scholars. No scholar is lazy. Some lazy people are students. All students are young.
Conclusions: I. No researcher is lazy. II. Some students are not scholars. III. Some young people are lazy.

Options:
- **(A)** Only I and II follow
- **(B)** Only II and III follow
- **(C)** All follow
- **(D)** None follows

Correct Answer: Option **(C)** — `All follow`

Detailed Step-by-Step Solution & Logical Derivation:
Researchers⊆scholars, and scholars∩lazy=∅, so researchers∩lazy=∅ (I). Some lazy are students, and since scholars∩lazy=∅, those lazy-students are not scholars — "some students are not scholars" (II). Those same lazy-students are also students, and all students are young, so those lazy-students are young — "some young people are lazy" (III). All three follow.

Question ITM-P05-Q03 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$5$_p$age$_0$1.png

Question Text:
Data Sufficiency. If x and y are positive integers, what is the remainder when x is divided by y?
I. x = 5y + 3
II. y = 7

Options:
- **(A)** Statement I alone is sufficient, but II alone is not

- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Even both statements together are not sufficient

Correct Answer: Option **(C)** — `Both statements together are sufficient, but neither alone is`

Detailed Step-by-Step Solution & Logical Derivation:
I alone: x=5y+3 only gives remainder 3 if y>3; for small y (e.g., y=2, x=13, 13÷2 leaves remainder 1, not 3) it fails — insufficient alone. II alone: y=7 with x unknown is insufficient. Together: x=5(7)+3=38, and 38÷7 leaves remainder 3 (valid since 3<7) — sufficient together.

Question ITM-P05-Q04 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$5$_p$age$_0$1.png

Question Text:
Data Sufficiency. In how many days can A and B complete a job together?
I. A is twice as efficient as B.
II. A alone can complete the job in 12 days.

Options:
- **(A)** Statement I alone is sufficient, but II alone is not
- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Even both statements together are not sufficient

Correct Answer: Option **(C)** — `Both statements together are sufficient, but neither alone is`

Detailed Step-by-Step Solution & Logical Derivation:
II alone gives only A's rate (1/12), with B's rate unknown. I alone gives only a ratio (2:1), with no absolute rate. Together: A's rate=1/12, B's rate=1/24 (half of A's), combined rate=1/12+1/24=3/24=1/8, so together they take 8 days — sufficient together.

Question ITM-P05-Q05 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$5$_p$age$_0$2.png

Question Text:
In a code, letters in odd positions of a word are shifted 2 places forward in the alphabet, and letters in even positions are shifted 1 place forward. Using this rule, SYSTEM is coded as UZUUGN. What is the code for SERVER?

Options:
- **(A)** UFTWGS
- **(B)** UFTWFS
- **(C)** UGTWGS
- **(D)** UFUWGS

Correct Answer: Option **(A)** — `UFTWGS`

Detailed Step-by-Step Solution & Logical Derivation:
S(odd,+2)=U. E(even,+1)=F. R(odd,+2)=T. V(even,+1)=W. E(odd,+2)=G. R(even,+1)=S. Code: UFTWGS.

Question ITM-P05-Q06 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$5$_p$age$_0$2.png

Question Text:
In a code: "data packet loss" = "zt rp qm", "loss causes retransmission" = "qm xy lp", "retransmission increases latency" = "lp wv ou", "latency affects data" = "ou nk zt", "packet causes latency" = "rp xy ou". What is the code for "increases affects packet"?

Options:
- **(A)** wv nk rp
- **(B)** wv nk qm
- **(C)** ou nk rp
- **(D)** wv xy rp

Correct Answer: Option **(A)** — `wv nk rp`

Detailed Step-by-Step Solution & Logical Derivation:
Pairwise comparison of the 5 statements resolves every word: data=zt, packet=rp, loss=qm, causes=xy, retransmission=lp, increases=wv, latency=ou, affects=nk. So "increases affects packet" = wv nk rp.

Question ITM-P05-Q07 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
A rectangular cuboid is cut by 2 parallel cuts along its length, 1 cut along its breadth, and 3 cuts along its height (each cut perpendicular to that dimension), producing smaller cuboids from an originally fully-painted surface. How many of the resulting small cuboids have exactly 1 painted face?

Options:
- **(A)** 4
- **(B)** 6
- **(C)** 8
- **(D)** 2

Correct Answer: Option **(A)** — `4`

Detailed Step-by-Step Solution & Logical Derivation:
With 2 cuts in length (3 pieces), 1 cut in breadth (2 pieces), 3 cuts in height (4 pieces), every piece touches an outer breadth-face (since there's no "middle" breadth layer). A piece has exactly 1 painted face only if it is interior in BOTH length and height (contributing 0 from each) while contributing exactly 1 from breadth (which is unavoidable). Interior length positions = 1 (out of 3), interior height positions = 2 (out of 4), and both breadth pieces qualify: 1×2×2 = 4.

Question ITM-P05-Q08 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
How many rectangles (of all sizes, including squares) are there in a 4×4 grid of unit squares?

Options:
- **(A)** 100
- **(B)** 90
- **(C)** 81
- **(D)** 110

Correct Answer: Option **(A)** — `100`

Detailed Step-by-Step Solution & Logical Derivation:
Choosing 2 of the 5 horizontal lines and 2 of the 5 vertical lines of a 4×4 grid gives a rectangle: C(5,2)×C(5,2) = 10×10 = 100.

Question ITM-P05-Q09 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$5$_p$age$_0$2.png

Question Text:
P and Q are a married couple. R is the son of P and Q. S is the daughter of P and Q. R is married to T. U is the son of R and T. V is the daughter of S. How is U related to V?

Options:
- **(A)** Cousins
- **(B)** Siblings
- **(C)** Nephew and Niece
- **(D)** Uncle and Niece

Correct Answer: Option **(A)** — `Cousins`

Detailed Step-by-Step Solution & Logical Derivation:
R and S are siblings (children of P and Q). U is R's son; V is S's daughter. Since their parents (R and S) are siblings, U and V are cousins.

Question ITM-P05-Q10 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$5$_p$age$_0$2.png

Question Text:
If A + B means A is the father of B, A − B means A is the mother of B, A × B means A is the son of B, and A ÷ B means A is the wife of B, which expression shows that M is the father-in-law of N (N being male)?

Options:
- **(A)** M + Q ÷ N
- **(B)** M ÷ Q + N
- **(C)** Q + M ÷ N
- **(D)** M × Q ÷ N

Correct Answer: Option **(A)** — `M + Q ÷ N`

Detailed Step-by-Step Solution & Logical Derivation:
M+Q: M is father of Q. Q÷N: Q is wife of N. So M is father of N's wife — M is N's father-in-law.

Question ITM-P05-Q11 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
A man starts from point O and walks 10 m North-East, then 10 m North-West, then 10 m South-West.
How far is he from O, and in which direction?

Visual Schematic / Logical Deduction Diagram:

```
(10m NW)
┌──────────────► [Point 2]
│ /
(10m NE) ▲ / (10m SW)
│ ▼
[Point O] ──► [FINISH]
Net position: returns to baseline axis at distance 10 m from O.
```

Visual Schematic / Logical Deduction Diagram:

```
(10m NW)
┌──────────────► [Point 2]
│ /
(10m NE) ▲ / (10m SW)
│ ▼
[Point O] ──► [FINISH]
Net position: returns to baseline axis at distance 10 m from O.
```

Visual Schematic / Logical Deduction Diagram:

```
(10m NW)
┌──────────────► [Point 2]
│ /
(10m NE) ▲ / (10m SW)
│ ▼
[Point O] ──► [FINISH]
Net position: returns to baseline axis at distance 10 m from O.
```

Visual Schematic / Logical Deduction Diagram:

```
(10m NW)
┌──────────────► [Point 2]
│ /
(10m NE) ▲ / (10m SW)
│ ▼
[Point O] ──► [FINISH]
Net position: returns to baseline axis at distance 10 m from O.
```

Options:
- **(A)** 10 m, North-West
- **(B)** 10 m, South-West
- **(C)** 14.14 m, North-West
- **(D)** 5 m, North-West

Correct Answer: Option **(A)** — `10 m, North-West`

Detailed Step-by-Step Solution & Logical Derivation:
Decomposing each 10 m leg into East/North components (each at 45°, so $\pm 5\sqrt{2}$ in each axis): totals work out to East$=-5\sqrt{2}$, North$=+5\sqrt{2}$, giving a resultant of $\sqrt{[(5\sqrt{2})^2+(5\sqrt{2})^2]}=\sqrt{100}=10$ m, in the North-West direction.

Question ITM-P05-Q12 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$5$_p$age$_0$2.png

Question Text:
A man is initially facing a bearing of 040°. After making exactly 2 equal clockwise turns, he ends up facing a bearing of 310°. What was the magnitude of each turn?

Options:
- **(A)** 135°
- **(B)** 142.5°
- **(C)** 120°
- **(D)** 150°

Correct Answer: Option **(A)** — `135°`

Detailed Step-by-Step Solution & Logical Derivation:
Total clockwise rotation needed = 310°−40° = 270°. Split equally over 2 turns: 270/2 = 135° each (verified: 40°+135°=175°, 175°+135°=310° ✓).

Question ITM-P05-Q13 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
Find the next term: 5, 14, 41, 122, 365, ?

Options:
- **(A)** 1094
- **(B)** 1095
- **(C)** 1093
- **(D)** 1096

Correct Answer: Option **(A)** — `1094`

Detailed Step-by-Step Solution & Logical Derivation:
Each term follows a(n) = 3×a(n−1) − 1: 122×3−1=365 ✓; next = 365×3−1 = 1094.

Question ITM-P05-Q14 🟢 [🔲 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** 🟢 [🔲 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 5 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
Find the next term in this 7-term series: 4, 9, 7, 16, 10, 25, 13, ?

Options:
- **(A)** 36
- **(B)** 16
- **(C)** 49
- **(D)** 30

Correct Answer: Option **(A)** — `36`

Detailed Step-by-Step Solution & Logical Derivation:
This series interleaves two sequences: odd positions (4,7,10,13,...) increase by 3; even positions (9,16,25,...) are 3^2,4^2,5^2. The 8th term (even position) continues the squares: 6^2 = 36.

Question ITM-P05-Q15 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 5 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$5$_p$age$_0$2.png

Question Text:
Find the next term: D, H, M, S, Z, ?

Options:
- **(A)** H
- **(B)** G
- **(C)** I
- **(D)** F

Correct Answer: Option **(A)** — `H`

Detailed Step-by-Step Solution & Logical Derivation:
Gaps increase by 1 each time: +4,+5,+6,+7; next gap +8. Z(26)+8=34, which wraps around (34−26=8) to H.

Question ITM-P06-Q01 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$6$_p$age$_0$1.png

Question Text:
Statements: Some bugs are features. All features are releases. No release is stable. All patches are stable.
Conclusions: I. Some bugs are not stable. II. No patch is a feature. III. Some releases are bugs. IV. Some stable things are bugs.

Options:
- **(A)** I and II only
- **(B)** II and III only
- **(C)** I, II and III only
- **(D)** All follow

Correct Answer: Option **(C)** — `I, II and III only`

Detailed Step-by-Step Solution & Logical Derivation:
Some bugs are features, features are releases, and no release is stable — so those bugs are not stable (I). Patches⊆stable and features∩stable=∅ (features are releases), so no patch is a feature (II). Some bugs are releases, which converts to "some releases are bugs" (III). IV does not follow: the only bugs we know are unstable; nothing says any bug is stable.

Question ITM-P06-Q02 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$6$_p$age$_0$1.png

Question Text:
Data Sufficiency. Is the product a × b positive?
I. a + b > 0.
II. a − b > 0.

Options:
- **(A)** Statement I alone is sufficient, but II alone is not
- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Even both statements together are not sufficient

Correct Answer: Option **(D)** — `Even both statements together are not sufficient`

Detailed Step-by-Step Solution & Logical Derivation:
a=3, b=1 satisfies both statements and gives a positive product; a=3, b=−1 also satisfies both (a+b=2>0, a−b=4>0) but gives a negative product. Even together they are not sufficient.

Question ITM-P06-Q03 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$6$_p$age$_0$1.png

Question Text:
Data Sufficiency. Is the positive integer n a perfect square?
I. n has exactly 3 positive divisors.
II. n is divisible by 9.

Options:
- **(A)** Statement I alone is sufficient, but II alone is not
- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Even both statements together are not sufficient

Correct Answer: Option **(A)** — `Statement I alone is sufficient, but II alone is not`

Detailed Step-by-Step Solution & Logical Derivation:
I: a number with exactly 3 divisors must be the square of a prime (p^2 has divisors 1, p, p^2), so n is a perfect square — sufficient. II: n=18 is divisible by 9 but is not a perfect square, while n=9 is — insufficient.

Question ITM-P06-Q04 ◯ [☐ PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [☐ PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$6$_p$age$_0$1.png

Question Text:
In a code, the 1st letter of a word is shifted 1 place forward, the 2nd letter 2 places forward, the 3rd letter 3 places forward, and so on. Using this rule, CODE is coded as DQGI. What is the code for LOGIC?

Options:
- **(A)** MQJMH
- **(B)** MPJLH
- **(C)** MQJNH
- **(D)** MQIMH

Correct Answer: Option **(A)** — `MQJMH`

Detailed Step-by-Step Solution & Logical Derivation:
L+1=M, O+2=Q, G+3=J, I+4=M, C+5=H, giving MQJMH (verified: the same rule turns CODE into DQGI).

Question ITM-P06-Q05 ◯ [☐ PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [☐ PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$6$_p$age$_0$1.png

Question Text:
In a code, the value of a word = (sum of the alphabetical positions of its letters × number of letters) −

(number of vowels in the word). Given that CAB has the value 17, what is the value of DIGIT?

Options:
- **(A)** 243
- **(B)** 245
- **(C)** 241
- **(D)** 247

Correct Answer: Option **(A)** — `243`

Detailed Step-by-Step Solution & Logical Derivation:
DIGIT: positions 4+9+7+9+20 = 49; 49×5 = 245; vowels (I, I) = 2; 245−2 = 243 (check: CAB = 6×3−1 = 17 ✓).

Question ITM-P06-Q06 `◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]`

> 🎯 **Exam Priority:** `◯ [🕮 PRACTICE VARIANT / REDUNDANT]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$6$_p$age$_0$2.png

Question Text:
A solid cube of side 7 units is painted on all faces and cut into 1-unit cubes. How many small cubes have at most 1 painted face?

Visual Schematic / Logical Deduction Diagram:

```
7 × 7 × 7 Cube (343 unit cubes)
Painted on all faces.
Exactly 1 face painted = 6 faces × (7 - 2)² = 6 × 25 = 150 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
7 × 7 × 7 Cube (343 unit cubes)
Painted on all faces.
Exactly 1 face painted = 6 faces × (7 - 2)² = 6 × 25 = 150 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
7 × 7 × 7 Cube (343 unit cubes)
Painted on all faces.
Exactly 1 face painted = 6 faces × (7 - 2)² = 6 × 25 = 150 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
7 × 7 × 7 Cube (343 unit cubes)
Painted on all faces.
Exactly 1 face painted = 6 faces × (7 - 2)² = 6 × 25 = 150 cubes.
```

Options:
- **(A)** 275
- **(B)** 125
- **(C)** 150
- **(D)** 335

Correct Answer: Option **(A)** — `275`

Detailed Step-by-Step Solution & Logical Derivation:
0 painted faces (interior) = 5^3 = 125; exactly 1 painted face = 6×5^2 = 150; total = 275.

Question ITM-P06-Q07 ◯ [☐ PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [☐ PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$6$_p$age$_0$2.png

Question Text:
How many rectangles that are NOT squares can be found in a 4 × 5 grid of unit squares?

Options:
- **(A)** 110
- **(B)** 150
- **(C)** 100
- **(D)** 120

Correct Answer: Option **(A)** — `110`

Detailed Step-by-Step Solution & Logical Derivation:
Total rectangles = C(5,2)×C(6,2) = 10×15 = 150. Squares = 20+12+6+2 = 40. Non-square rectangles = 150−40 = 110.

Question ITM-P06-Q08 ◯ [☐ PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [☐ PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$6$_p$age$_0$2.png

Question Text:
If P # Q means P is the mother of Q, P $ Q means P is the brother of Q, and P @ Q means P is the wife of Q, then in the expression A $ B # C @ D, how is A related to D?

Options:
- **(A)** Wife's maternal uncle (uncle-in-law)
- **(B)** Brother-in-law
- **(C)** Father-in-law
- **(D)** Maternal uncle

Correct Answer: Option **(A)** — `Wife's maternal uncle (uncle-in-law)`

Detailed Step-by-Step Solution & Logical Derivation:
A is the brother of B, and B is the mother of C, so A is C's maternal uncle. C is the wife of D, so A is D's wife's maternal uncle — an uncle-in-law.

Question ITM-P06-Q09 ◯ [☐ PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [☐ PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_06_page_02.png

Question Text:
A is 20 km due North of B. C is 15 km due East of B. What is the bearing of A from C (measured clockwise from North, to the nearest degree)?

Visual Schematic / Logical Deduction Diagram:

```
A (North)
▲
| (20 km)
|
B ————————► C (East, 15 km)
Triangle ABC is a right-angled triangle at B.
tan(∠ACB) = AB / BC = 20 / 15 = 4/3 => ∠ACB ≈ 53.13°
Bearing of A from C = 360° - (90° + 53.13°) = 307° (or N 53° W).
```

Visual Schematic / Logical Deduction Diagram:

```
A (North)
▲
| (20 km)
|
B ————————► C (East, 15 km)
Triangle ABC is a right-angled triangle at B.
tan(∠ACB) = AB / BC = 20 / 15 = 4/3 => ∠ACB ≈ 53.13°
Bearing of A from C = 360° - (90° + 53.13°) = 307° (or N 53° W).
```

Visual Schematic / Logical Deduction Diagram:

```
A (North)
▲
| (20 km)
|
B ————————► C (East, 15 km)
Triangle ABC is a right-angled triangle at B.
tan(∠ACB) = AB / BC = 20 / 15 = 4/3 => ∠ACB ≈ 53.13°
Bearing of A from C = 360° - (90° + 53.13°) = 307° (or N 53° W).
```

Visual Schematic / Logical Deduction Diagram:


```
A (North)
▲
| (20 km)
|
B ————————► C (East, 15 km)
Triangle ABC is a right-angled triangle at B.
tan(∠ACB) = AB / BC = 20 / 15 = 4/3 => ∠ACB ≈ 53.13°
Bearing of A from C = 360° - (90° + 53.13°) = 307° (or N 53° W).
```


Options:
- **(A)** 323°
- **(B)** 307°
- **(C)** 037°
- **(D)** 217°

Correct Answer: Option **(A)** — `323°`

Detailed Step-by-Step Solution & Logical Derivation:
Taking East and North as axes, A is at (0,20), B at (0,0), C at (15,0). The vector from C to A is (−15, 20): 36.87° west of north, giving a bearing of 360°−36.87° ≈ 323°.

Question ITM-P06-Q10 ◯ [☐ PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [☐ PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper₀6ₚage₀2.png

Question Text:
Five people A, B, C, D and E stand in a queue. A is ahead of B but behind C. D stands immediately behind B. E is ahead of C. Who stands third from the front?

Options:
- **(A)** A
- **(B)** B
- **(C)** C
- **(D)** E

Correct Answer: Option **(A)** — `A`

Detailed Step-by-Step Solution & Logical Derivation:
E is ahead of C, C ahead of A, A ahead of B, and D immediately behind B: E, C, A, B, D. Third from the front is A.

Question ITM-P06-Q11 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$6$_p$age$_0$2.png

Question Text:
Find the next term: 3, 5, 11, 29, 83, ?

Options:
- **(A)** 245
- **(B)** 247
- **(C)** 239
- **(D)** 251

Correct Answer: Option **(A)** — `245`

Detailed Step-by-Step Solution & Logical Derivation:
Differences are 2, 6, 18, 54 (each ×3); the next difference is 162. 83+162 = 245.

Question ITM-P06-Q12 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$6$_p$age$_0$2.png

Question Text:
Find the next term: 1, 4, 27, 16, 125, 36, ?

Options:
- **(A)** 343
- **(B)** 49
- **(C)** 216
- **(D)** 64

Correct Answer: Option **(A)** — `343`

Detailed Step-by-Step Solution & Logical Derivation:
Odd positions are cubes (1^3, 3^3, 5^3) and even positions are squares (2^2, 4^2, 6^2). The 7th term is 7^3 = 343.

Question ITM-P06-Q13 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$6$_p$age$_0$2.png

Question Text:
Find the next group: ACF, GIL, MOR, ?

Options:
- **(A)** SUX
- **(B)** SVY
- **(C)** TVY
- **(D)** SUW

Correct Answer: Option **(A)** — `SUX`

Detailed Step-by-Step Solution & Logical Derivation:
Each group has three letters with gaps +2, +3 (A→C→F, G→I→L, M→O→R), and each group starts 6 letters after the previous one. The next group is S, U, X = SUX.

Question ITM-P06-Q14 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$6$_p$age$_0$2.png

Question Text:
A square sheet of paper is folded in half three times (each fold halves the previous folded shape). A single hole is then punched through all layers. When the sheet is fully unfolded, how many holes does it have?

Options:
- **(A)** 8
- **(B)** 4
- **(C)** 6
- **(D)** 16

Correct Answer: Option **(A)** — `8`

Detailed Step-by-Step Solution & Logical Derivation:
Each fold doubles the layers: $2^3 = 8$ layers, so one punch makes 8 holes.

Question ITM-P06-Q15 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 6 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$6$_p$age$_0$2.png

Question Text:
Data Sufficiency. How is P related to Q?
I. P is the daughter of R.
II. R's brother is Q's father.

Options:
- **(A)** Statement I alone is sufficient, but II alone is not
- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Even both statements together are not sufficient

Correct Answer: Option **(C)** — `Both statements together are sufficient, but neither alone is`

Detailed Step-by-Step Solution & Logical Derivation:
I alone gives only P's parent; II alone links R and Q's father without involving P. Together: P is the daughter of R, and R is the sibling of Q's father, so P and Q are cousins.

Question ITM-P07-Q01 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$7$_p$age$_0$1.png

Question Text:
Statements: Some servers are clients. All clients are nodes. No node is offline. Some offline things are routers.
Conclusions: I. Some servers are not offline. II. Some routers are not nodes. III. No client is a router. IV. Some nodes are servers.

Options:
- **(A)** I and II only
- **(B)** I, II and IV only
- **(C)** I, II, III and IV
- **(D)** II, III and IV only

Correct Answer: Option **(B)** — `I, II and IV only`

Detailed Step-by-Step Solution & Logical Derivation:
Some servers are clients and clients are nodes, which are never offline, so those servers are not offline (I). Some offline things are routers, and nodes are never offline, so those routers are not nodes (II). Some servers are nodes, which converts to "some nodes are servers" (IV). III does not follow, because other routers (not the offline ones) might be clients.

Question ITM-P07-Q02 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper$_0$7$_p$age$_0$1.png

Question Text:
Data Sufficiency. What is the value of x + y?
I. $x^2 + 2xy + y^2 = 49$
II. x and y are both positive.

Options:
- **(A)** Statement I alone is sufficient, but II alone is not
- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Even both statements together are not sufficient

Correct Answer: Option **(C)** — `Both statements together are sufficient, but neither alone is`

Detailed Step-by-Step Solution & Logical Derivation:
I gives $(x + y)^2 = 49$, so $x + y = 7$ or -7. II alone gives no value. Together, $x + y$ must be positive, so $x + y = 7$.

Question ITM-P07-Q03 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 2

Visual Reference Plate: paper_07_page_02.png

Question Text:
Data Sufficiency. Is the average of the set S greater than 10?
I. The median of S is 12.
II. S contains exactly 5 positive integers.

Options:
- **(A)** Statement I alone is sufficient, but II alone is not
- **(B)** Statement II alone is sufficient, but I alone is not
- **(C)** Both statements together are sufficient, but neither alone is
- **(D)** Even both statements together are not sufficient

Correct Answer: Option **(D)** — `Even both statements together are not sufficient`

Detailed Step-by-Step Solution & Logical Derivation:
A set such as {1, 1, 12, 12, 12} has median 12 and 5 positive integers but an average of 7.6, while {12, 12, 12, 13, 100} has an average above 10. Even together, the statements are not sufficient.

Question ITM-P07-Q04 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_07_page_02.png

Question Text:
In a code, each letter is replaced by its opposite letter in the alphabet (A↔Z, B↔Y, C↔X, and so on), and the resulting word is then written in reverse order. Using this rule, CODE is coded as VWLX. What is the code for SYSTEM?

Options:
- **(A)** NVGHBH
- **(B)** HBHGVN
- **(C)** NVGHBG
- **(D)** MVGHBH

Correct Answer: Option **(A)** — `NVGHBH`

Detailed Step-by-Step Solution & Logical Derivation:
S→H, Y→B, S→H, T→G, E→V, M→N gives HBHGVN, which reversed is NVGHBH (check: CODE→XLWV→VWLX ✓).

Question ITM-P07-Q05 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`

> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_07_page_02.png

Question Text:
In a certain code, ROSE is written as 6821, CHAIR is written as 73456, and PREACH is written as 961473. How is SEARCH written in that code?

Options:
- **(A)** 214673
- **(B)** 214763
- **(C)** 241673
- **(D)** 214637

Questions 6 and 7 are based on the following information.
Eight people A, B, C, D, E, F, G and H sit around a circular table facing the centre. A sits opposite E. B sits immediately to the right of A. C sits second to the left of B. D sits immediately to the right of E. G sits opposite C. F sits second to the right of A.

Correct Answer: Option **(A)** — `214673`

Detailed Step-by-Step Solution & Logical Derivation:
From the three words: R=6, O=8, S=2, E=1, C=7, H=3, A=4, I=5, P=9. SEARCH = S(2) E(1) A(4) R(6) C(7) H(3) = 214673.

Question ITM-P07-Q06 `◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]`

> 🎯 **Exam Priority:** `◯ [🕮 PRACTICE VARIANT / REDUNDANT]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_07_page_02.png

Question Text:
Who sits opposite B?

Options:
- **(A)** D
- **(B)** H
- **(C)** F
- **(D)** G

Correct Answer: Option **(A)** — `D`

Detailed Step-by-Step Solution & Logical Derivation:
Fix A at one seat and number the seats clockwise from A: A0, C1, H2, D3, E4, G5, F6, B7. (E is opposite A at 4; B is to A's right at 7; C is second to B's left at 1; D is to E's right at 3; G is opposite C at 5; F is second to A's right at 6; H takes the last seat.) The seat opposite B (7) is seat 3, which is D.

Question ITM-P07-Q07 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_07_page_02.png

Question Text:
Who sits third to the left of F?

Options:
- **(A)** C
- **(B)** A
- **(C)** H
- **(D)** B

Questions 8 and 9 are based on the following table.
Year
Revenue (₹ crore)
Expenditure (₹ crore)
2021
420
350
2022
510
400
2023
600
480
2024
660
550
2025
780
600

(Profit = Revenue − Expenditure.)

Correct Answer: Option **(A)** — `C`

Detailed Step-by-Step Solution & Logical Derivation:
F is at seat 6. Moving to the left (clockwise): first is B (7), second is A (0), third is C (1).

Question ITM-P07-Q08 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`

> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper07page02.png

Question Text:
Over the five years together, total profit is what percentage of total expenditure (to the nearest whole number)?

Options:
- **(A)** 25%
- **(B)** 24%
- **(C)** 26%
- **(D)** 28%

Correct Answer: Option **(A)** — `25%`

Detailed Step-by-Step Solution & Logical Derivation:
Profits: 70, 110, 120, 110, 180, totalling 590. Total expenditure = 350+400+480+550+600 = 2380. 590/2380 ≈ 24.8%, which is 25%.

Question ITM-P07-Q09 ◯ [☐ PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [☐ PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper07page02.png

Question Text:
Between which two consecutive years was the percentage growth in revenue the lowest?

Options:
- **(A)** 2021–2022
- **(B)** 2022–2023
- **(C)** 2023–2024
- **(D)** 2024–2025

Correct Answer: Option **(C)** — `2023–2024`

Detailed Step-by-Step Solution & Logical Derivation:
Growth: 2021–22 = 21.4%, 2022–23 = 17.6%, 2023–24 = 10%, 2024–25 = 18.2%. The lowest is 2023–2024.

Question ITM-P07-Q10 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper$_0$7$_p$age$_0$2.png

Question Text:
A is the mother of B. B is the sister of C. C is the father of D. E is the only brother of A. How is E related to D?

Options:
- **(A)** Paternal grand-uncle
- **(B)** Maternal uncle
- **(C)** Paternal uncle
- **(D)** Grandfather

Correct Answer: Option **(A)** — `Paternal grand-uncle`

Detailed Step-by-Step Solution & Logical Derivation:
A is the mother of B, and B is C's sister, so A is also C's mother. C is D's father, so A is D's paternal grandmother. E is A's only brother, so E is D's paternal grand-uncle.

Question ITM-P07-Q11 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 3
Visual Reference Plate: paper$_0$7$_p$age$_0$3.png

Question Text:
A ship sails 20 km on a bearing of 030°, and then 20 km on a bearing of 330°. What is its final position relative to the starting point?

Options:
- **(A)** 20√3 km (about 34.6 km) due North
- **(B)** 20 km due North
- **(C)** 40 km due North
- **(D)** 20√3 km due East

Correct Answer: Option **(A)** — `20√3 km (about 34.6 km) due North`

Detailed Step-by-Step Solution & Logical Derivation:
Leg 1: East = 20 sin30° = 10, North = 20 cos30° = 10√3. Leg 2 (bearing 330°): East = −10, North =

+10√3. The East parts cancel; the North parts add to 20√3 ≈ 34.6 km, so the ship is 20√3 km due North.

Question ITM-P07-Q12 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 3
Visual Reference Plate: paper$_0$7$_p$age$_0$3.png

Question Text:
Find the next term: 1, 3, 12, 60, 360, ?

Options:
- **(A)** 2520
- **(B)** 2160
- **(C)** 1800
- **(D)** 2400

Correct Answer: Option **(A)** — `2520`

Detailed Step-by-Step Solution & Logical Derivation:
Each term is the previous term multiplied by 3, 4, 5, 6 and then 7: 360 × 7 = 2520.

Question ITM-P07-Q13 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 3
Visual Reference Plate: paper$_0$7$_p$age$_0$3.png

Question Text:
Find the next term: 3, 8, 6, 14, 12, 26, ?

Options:
- **(A)** 24
- **(B)** 22
- **(C)** 50
- **(D)** 52

Correct Answer: Option **(A)** — `24`

Detailed Step-by-Step Solution & Logical Derivation:
Two interleaved series. Odd positions: 3, 6, 12, then ×2 gives 24. Even positions: 8, 14, 26 (each is double the previous minus 2). The 7th term, in an odd position, is 24.

Question ITM-P07-Q14 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 3
Visual Reference Plate: paper$_0$7$_p$age$_0$3.png

Question Text:
The image of a wall clock in a plane mirror shows the time as 8:25. What is the actual time?

Options:
- **(A)** 3:35
- **(B)** 4:35
- **(C)** 3:25
- **(D)** 4:25

Correct Answer: Option **(A)** — `3:35`

Detailed Step-by-Step Solution & Logical Derivation:
Actual time = 11:60 − 8:25 = 3:35.

Question ITM-P07-Q15 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(Infosys System Engineer ITM Mock Paper 7 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 3
Visual Reference Plate: paper$_0$7$_p$age$_0$3.png

Question Text:
In a row of 40 students, A is 15th from the left and B is 18th from the right. How many students are between A and B?

Options:
- **(A)** 7
- **(B)** 6
- **(C)** 8
- **(D)** 9

Correct Answer: Option **(A)** — `7`

Detailed Step-by-Step Solution & Logical Derivation:
B is 18th from the right, so B is 40−18+1 = 23rd from the left. Students between A (15th) and B (23rd) = 23−15−1 = 7.
